EART 22201 Igneous minerals and processes

Intraplate volcanism

Lesson 7



margaret.hartley@manchester.ac.uk

Intraplate volcanism

1.1 Present-day locations of intraplate volcanism

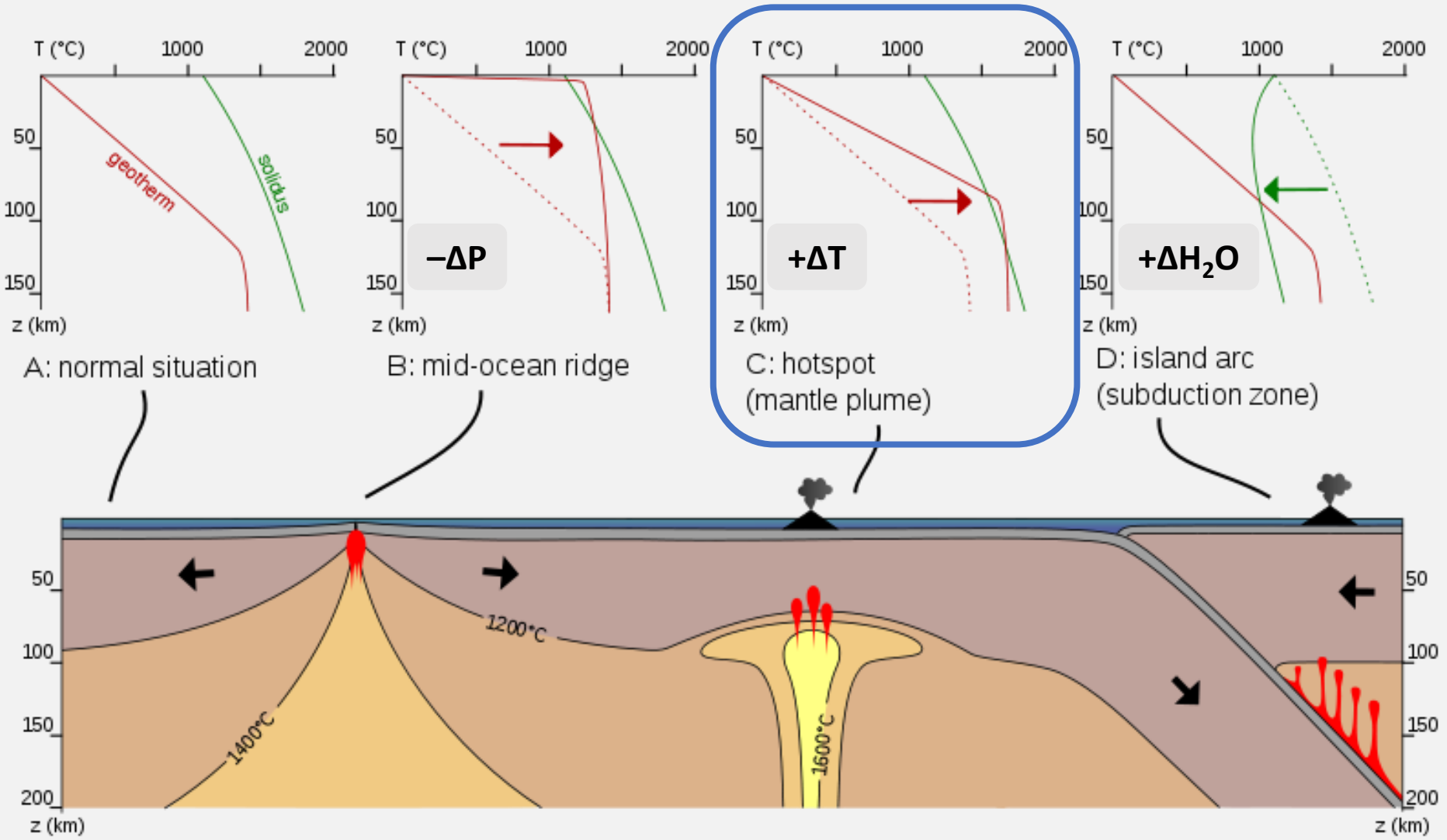
1.2 Active upwelling

1.3 Mantle temperature and melt production



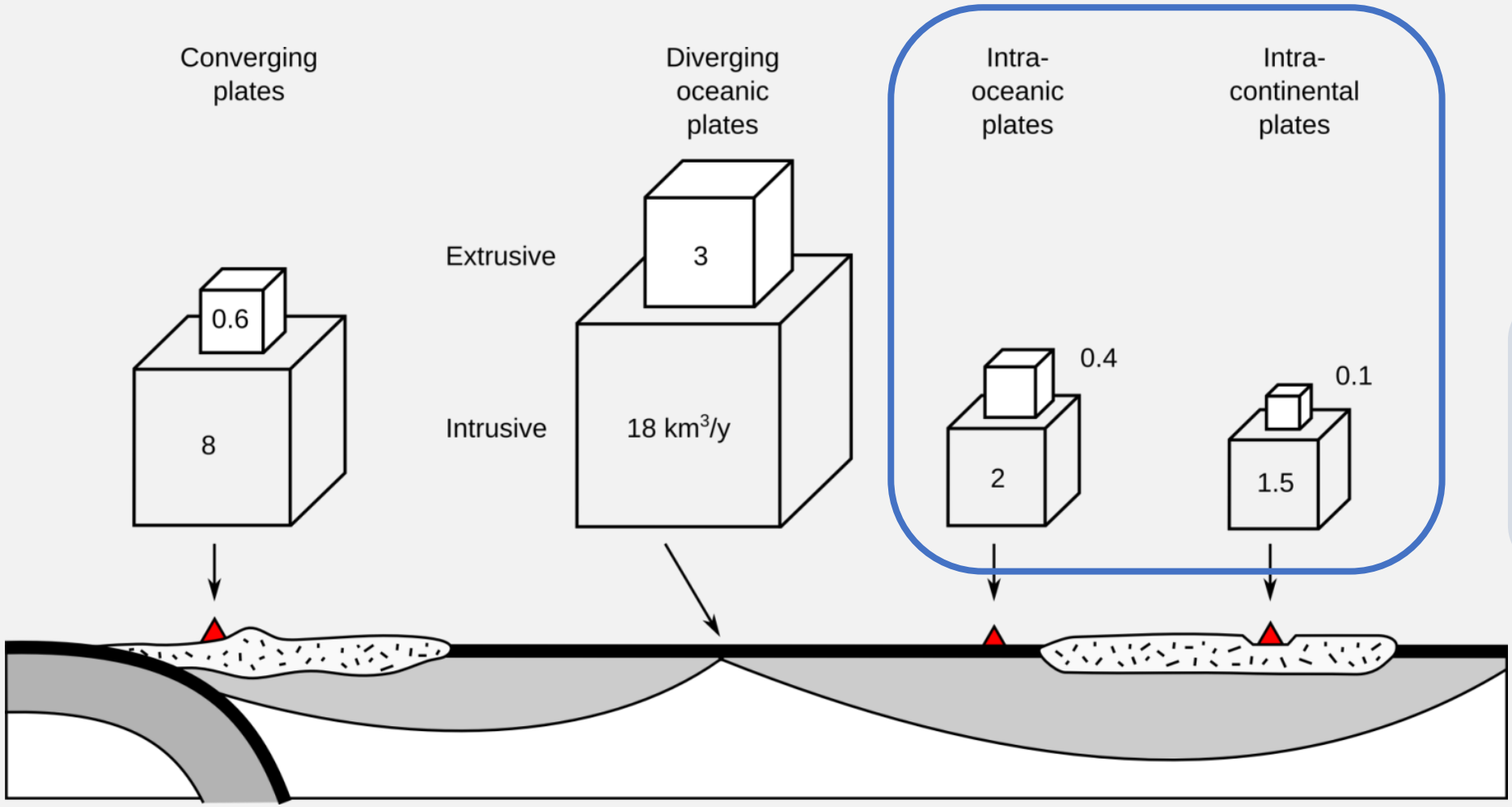
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Where and how are different primary melt compositions produced?



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A global inventory of magma production

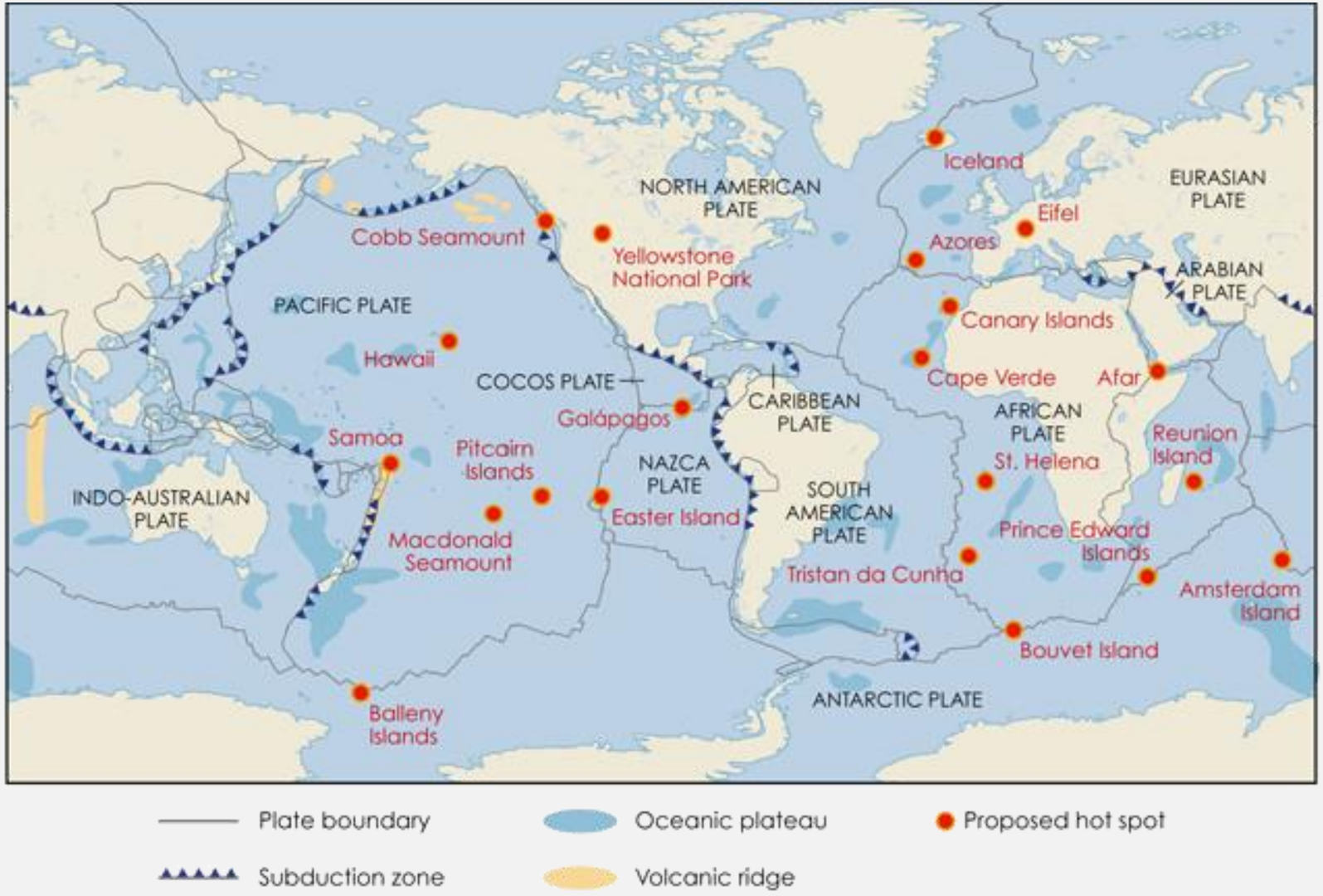


Intraplate magmatism:

- ~10% of global extrusive magma volume
- ~10% of global intrusive magma volume

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Intraplate volcanism: mantle plumes and hotspots



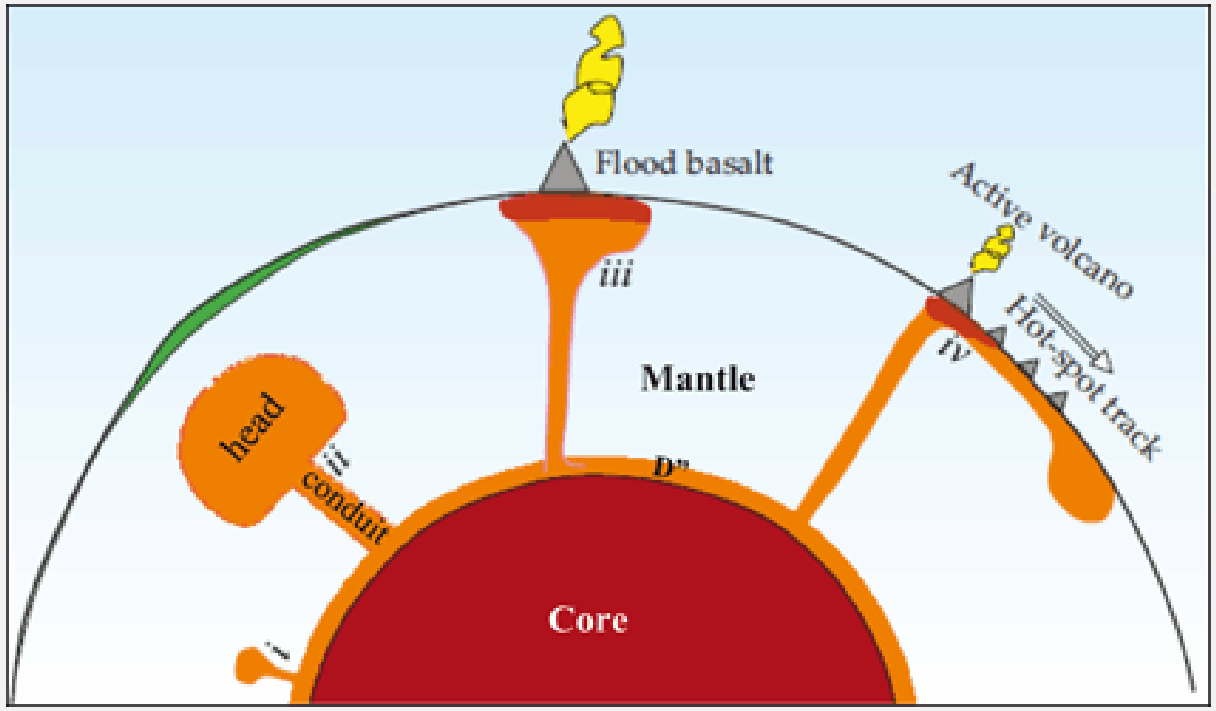
Many instances of intraplate volcanism are attributed to mantle plumes, or hotspots, in the Earth's mantle

Intraplate volcanism can also be caused by:

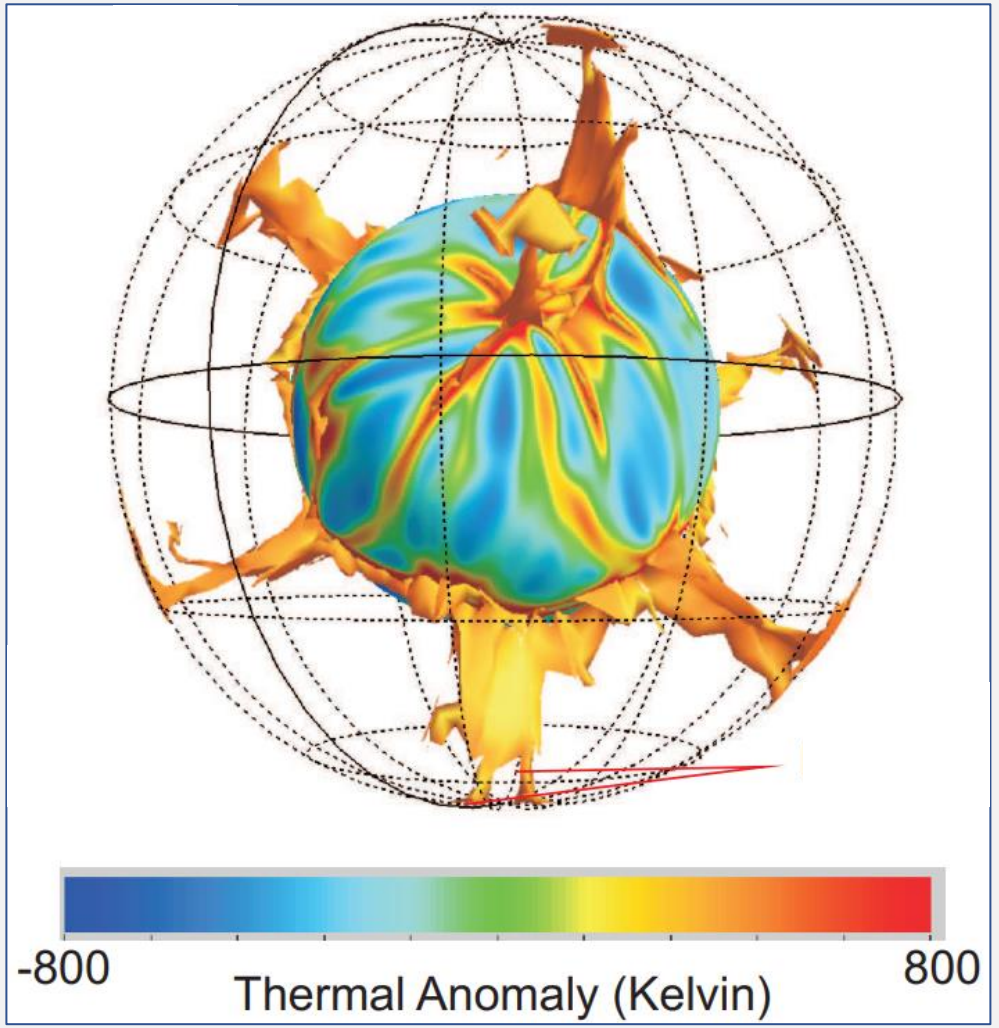
- Lithospheric extension within tectonic plates
- Impact events

margaret.hartley@manchester.ac.uk

Mantle plumes: thermal convection in the silicate Earth



- Buoyant upwelling of hot asthenospheric mantle
- At the base of the lithosphere the hot material spreads laterally
- Motion of tectonic plate over the plume head may generate hotspot tracks



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Intraplate volcanism

1.1 Present-day locations of intraplate volcanism

1.2 Active upwelling

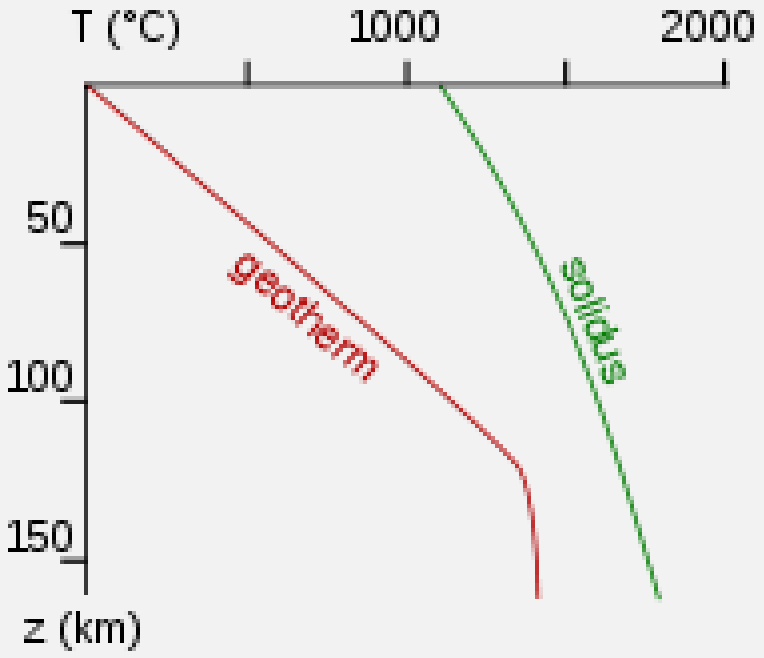
1.3 Mantle temperature and melt production



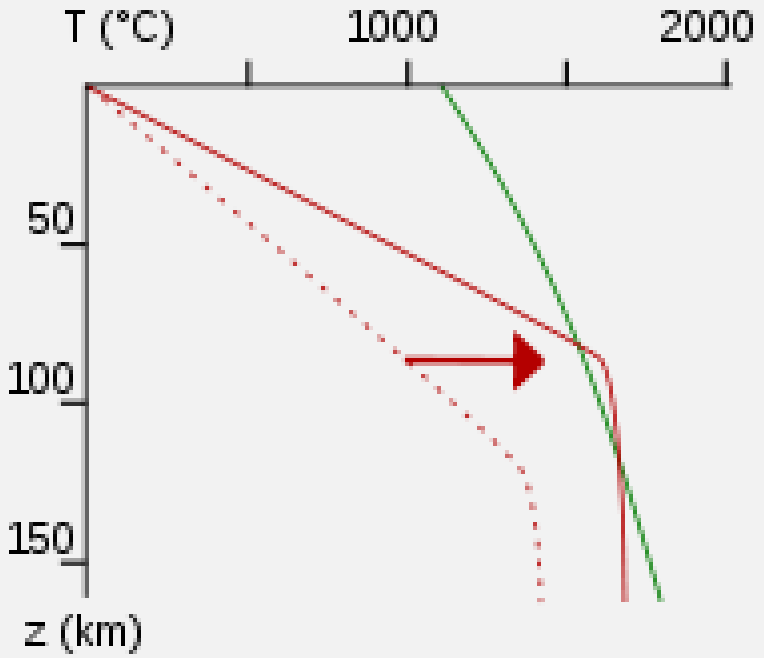
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Active upwelling in intraplate settings

Ambient mantle geotherm



Mantle plume geotherm



Active upwelling

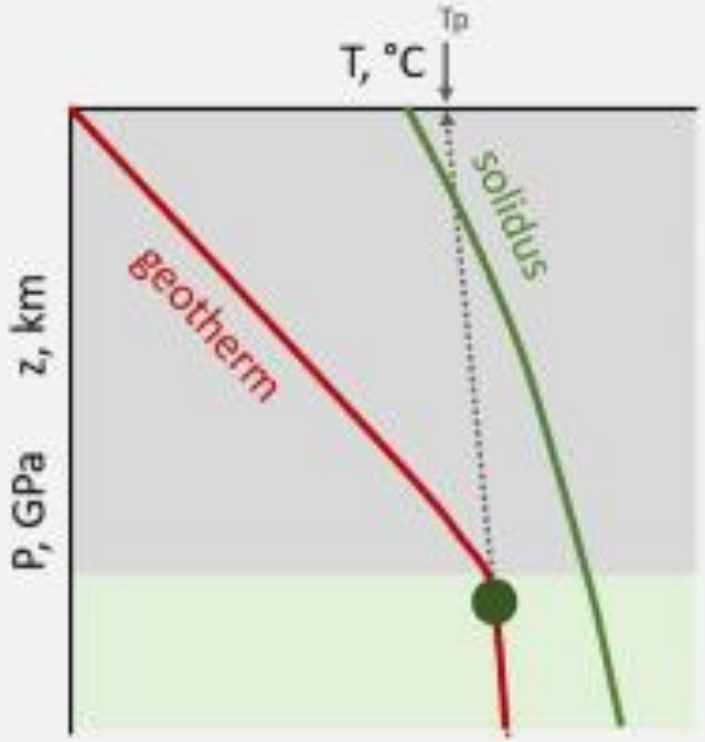
Deep melting caused by elevated mantle potential temperature.

High Tp implies the presence of a hot, convecting mantle plume.

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Active upwelling in intraplate settings



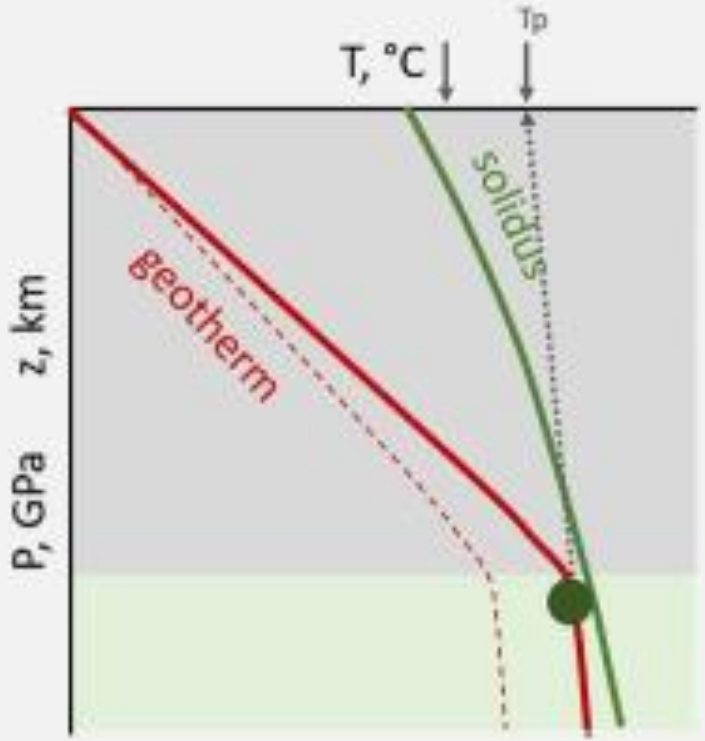


Upwelling mantle intersects the dry solidus at depth due to elevated mantle potential temperature. Melting is deeper than at mid-ocean ridges.

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Active upwelling in intraplate settings

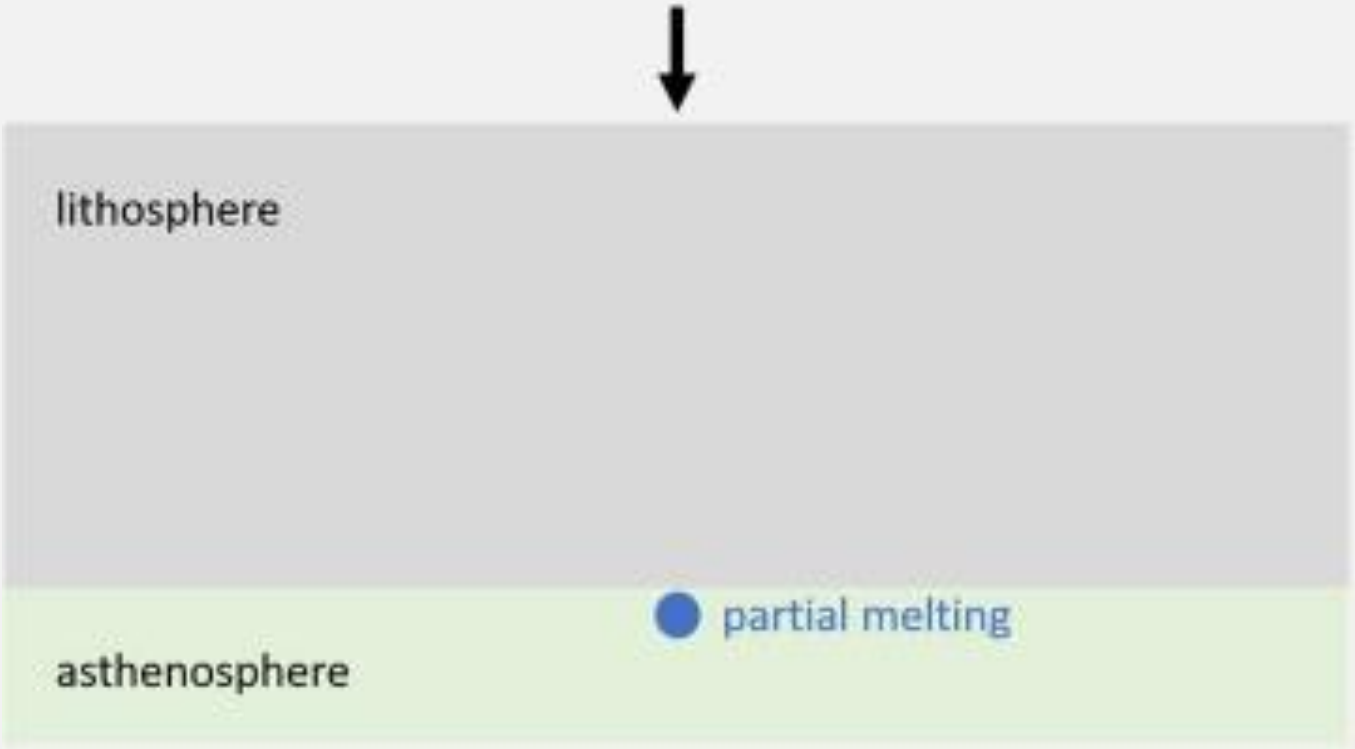


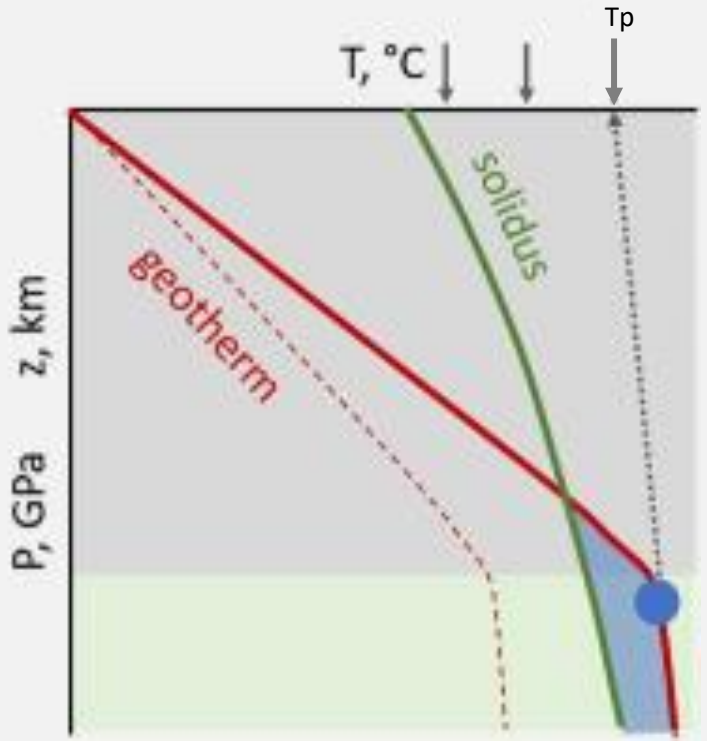


Upwelling mantle intersects the dry solidus at depth due to elevated mantle potential temperature. Melting is deeper than at mid-ocean ridges.

margaret.hartley@manchester.ac.uk

Active upwelling in intraplate settings





Upwelling mantle intersects the dry solidus at depth due to elevated mantle potential temperature. Melting is deeper than at mid-ocean ridges.

margaret.hartley@manchester.ac.uk

Intraplate volcanism

1.1 Present-day locations of intraplate volcanism

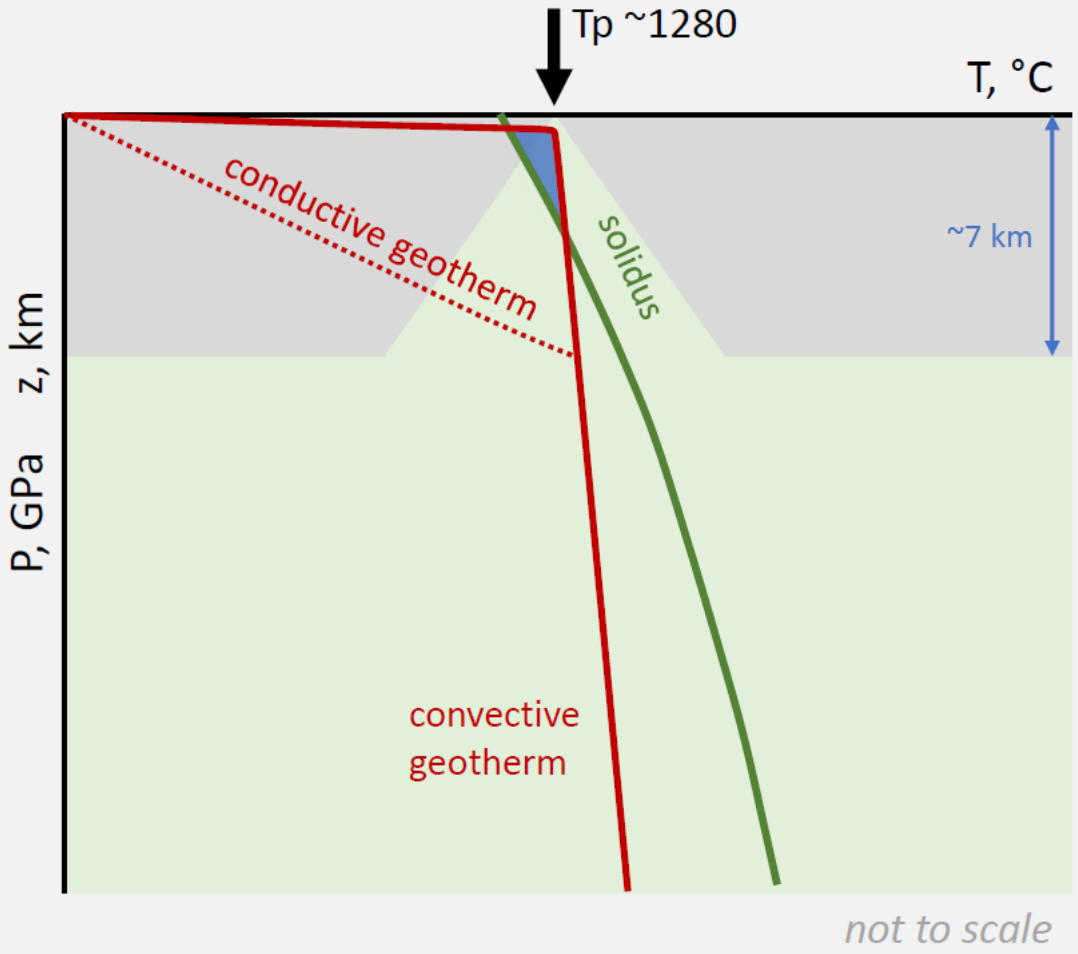
1.2 Active upwelling

1.3 Mantle temperature and melt production



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Tp and lithospheric thickness



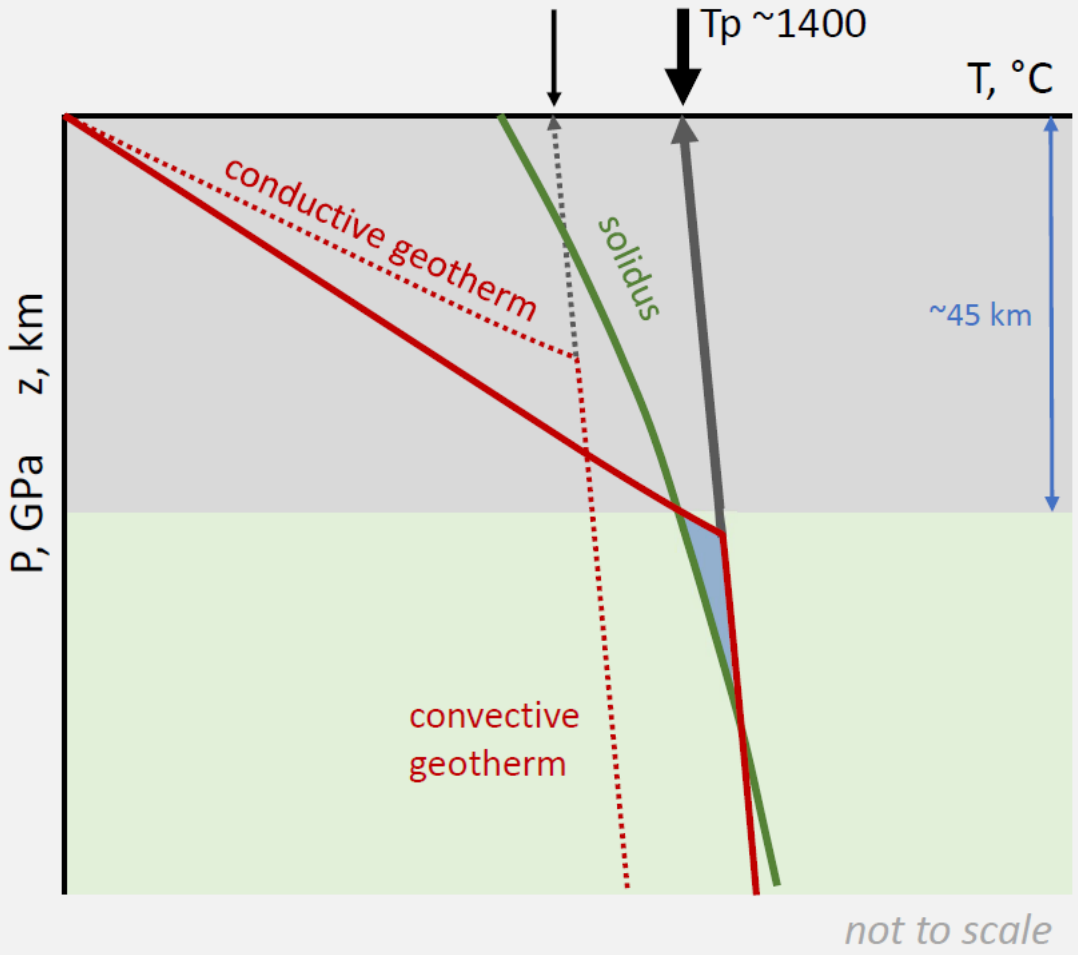
Ambient mantle Tp = 1280 °C

Iceland mantle Tp ≈ 1400 °C

Hawaii mantle Tp ≈ 1550 °C

margaret.hartley@manchester.ac.uk

Tp and lithospheric thickness



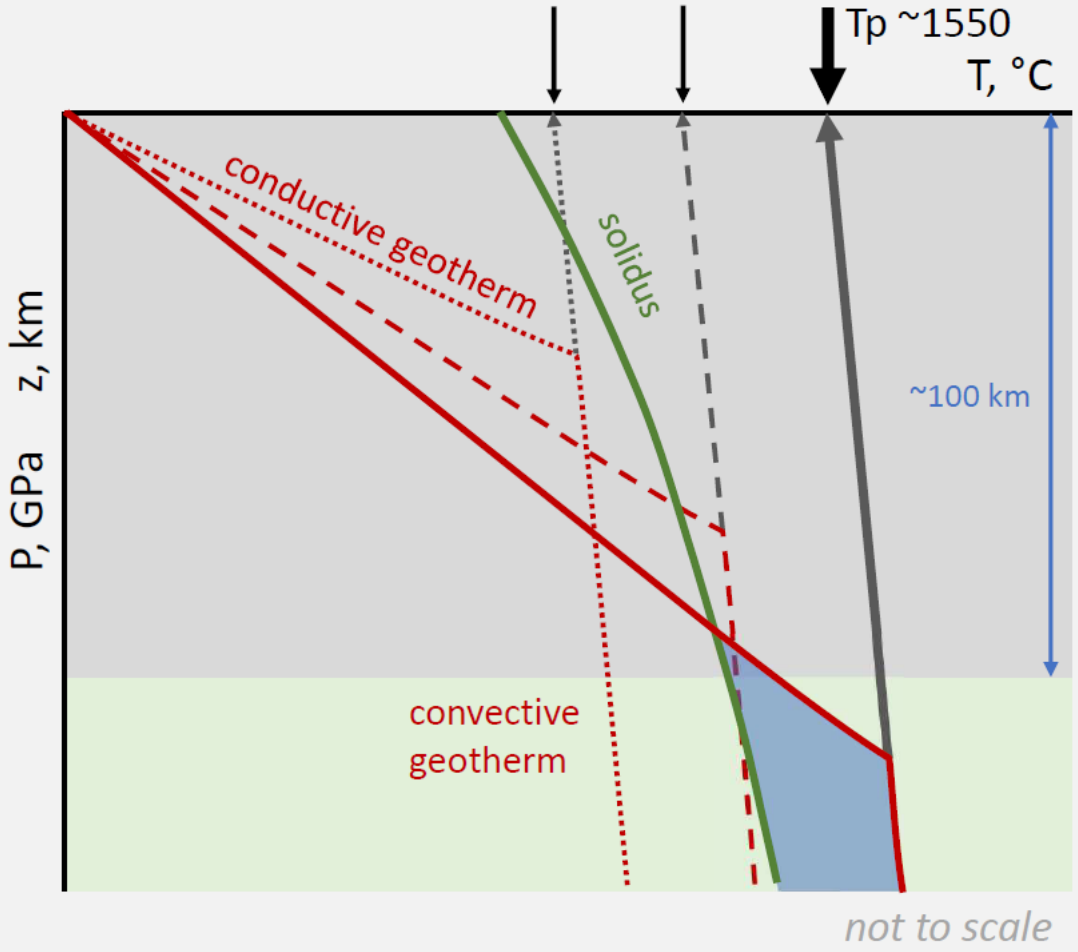
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margaret.hartley@manchester.ac.uk

Tp and lithospheric thickness



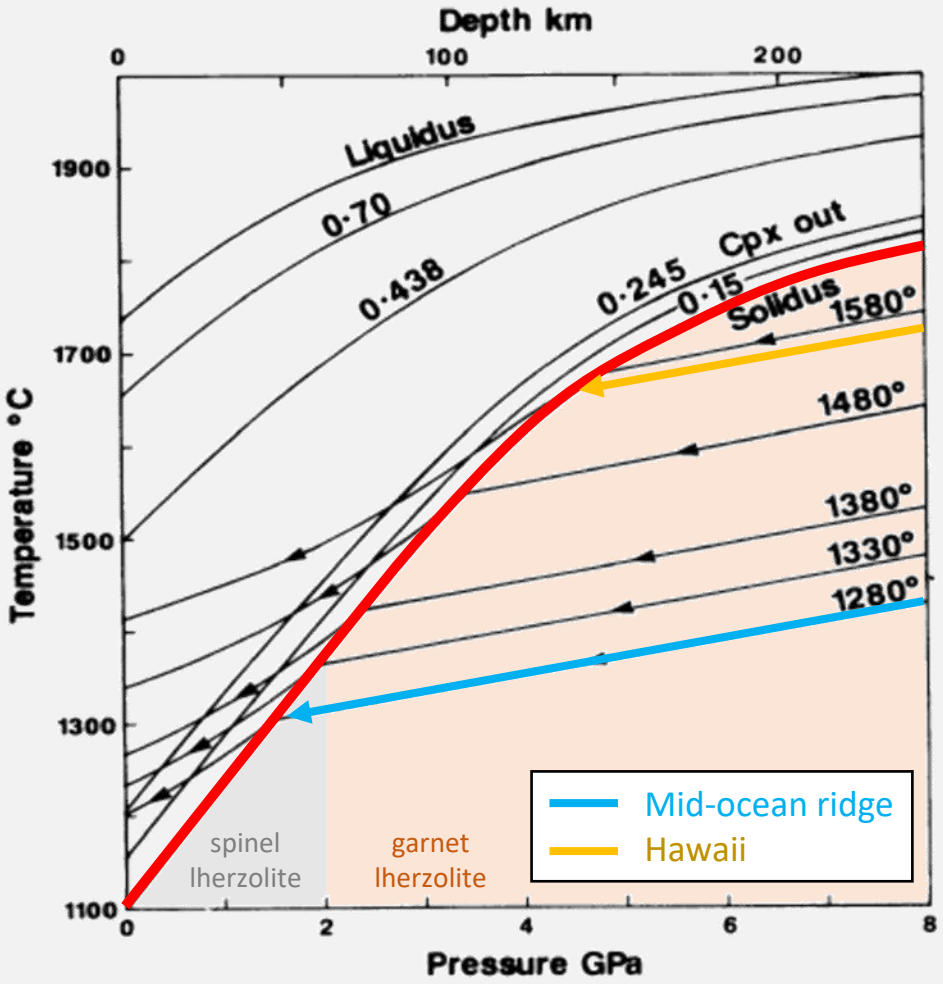
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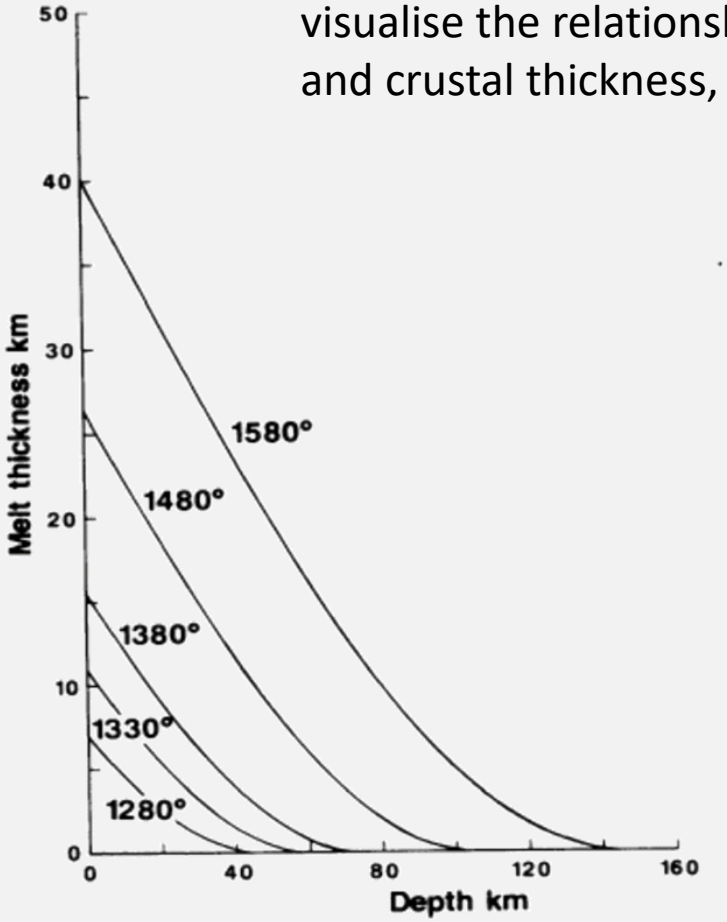
Iceland mantle Tp ≈ 1400 °C

Hawaii mantle Tp ≈ 1550 °C

margaret.hartley@manchester.ac.uk

Mantle temperature and melt production





These diagrams are included in case you find it helpful to visualise the relationships between Tp, depth of melting, and crustal thickness, in a slightly different way.

Left: Red line is the solidus where melting begins. Straight lines show the geothermal gradient for different mantle potential temperatures. The higher the Tp, the deeper the solidus is crossed. Therefore, high Tp means that melting commences at high pressure.

Right: The x axis is the pressure that melting begins. The y axis corresponds to the thickness of crust produced once all the melts have solidified. High Tp means a deeper onset of melting, and a greater melt thickness.

margaret.hartley@manchester.ac.uk

Intraplate volcanism

You should now be able to:

- Explain the concept of active upwelling and its link to mantle potential temperature
- Construct a P-T diagram to show how the geotherm and solidus intersect to cause melting in intraplate settings
- Qualitatively describe the relationship between mantle potential temperature, depth of melting, and melt thickness



margaret.hartley@manchester.ac.uk

Intraplate volcanism

Oceanic intraplate volcanism: Ocean islands. Case study: Hawaii

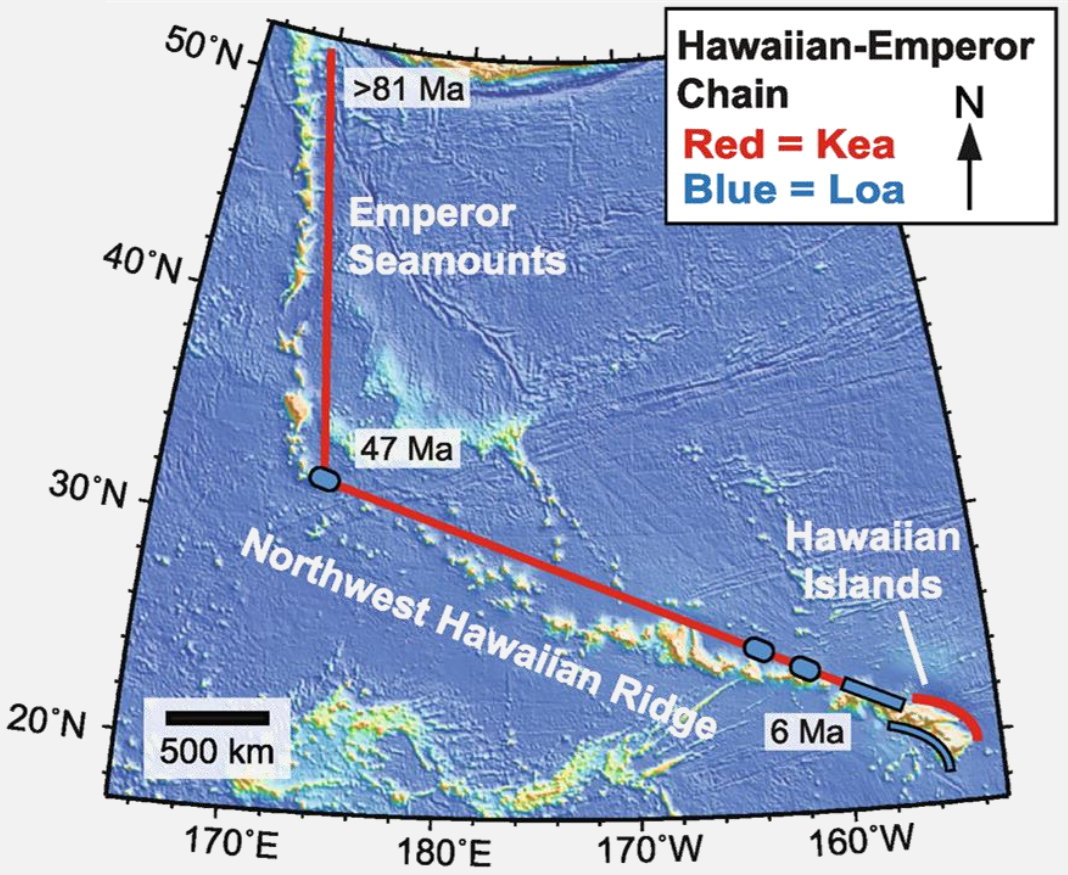
2.1 Lifecycle of Hawaiian volcanoes

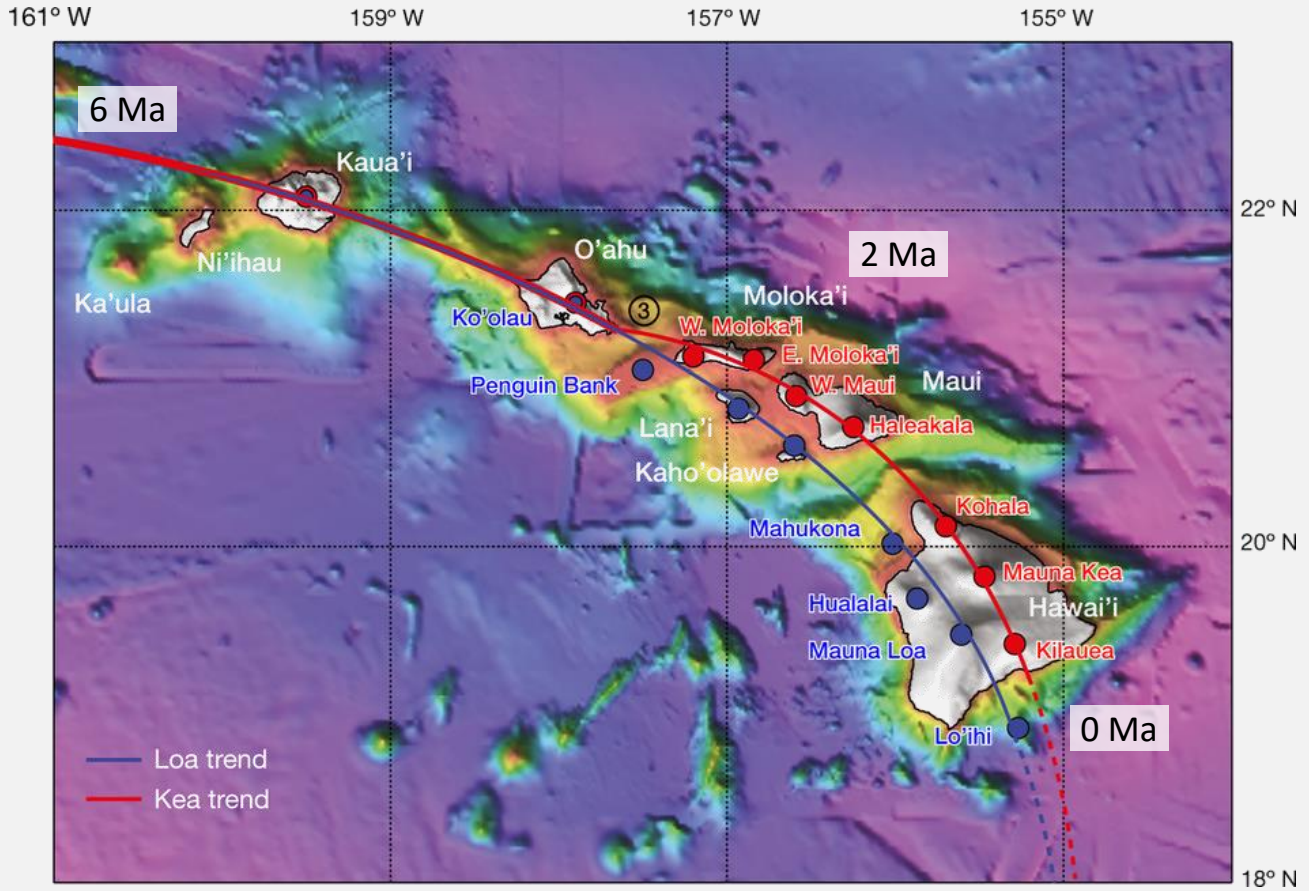
2.2 Melt production over the Hawaii mantle plume



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The Hawaii-Emperor Chain and Hawaiian islands

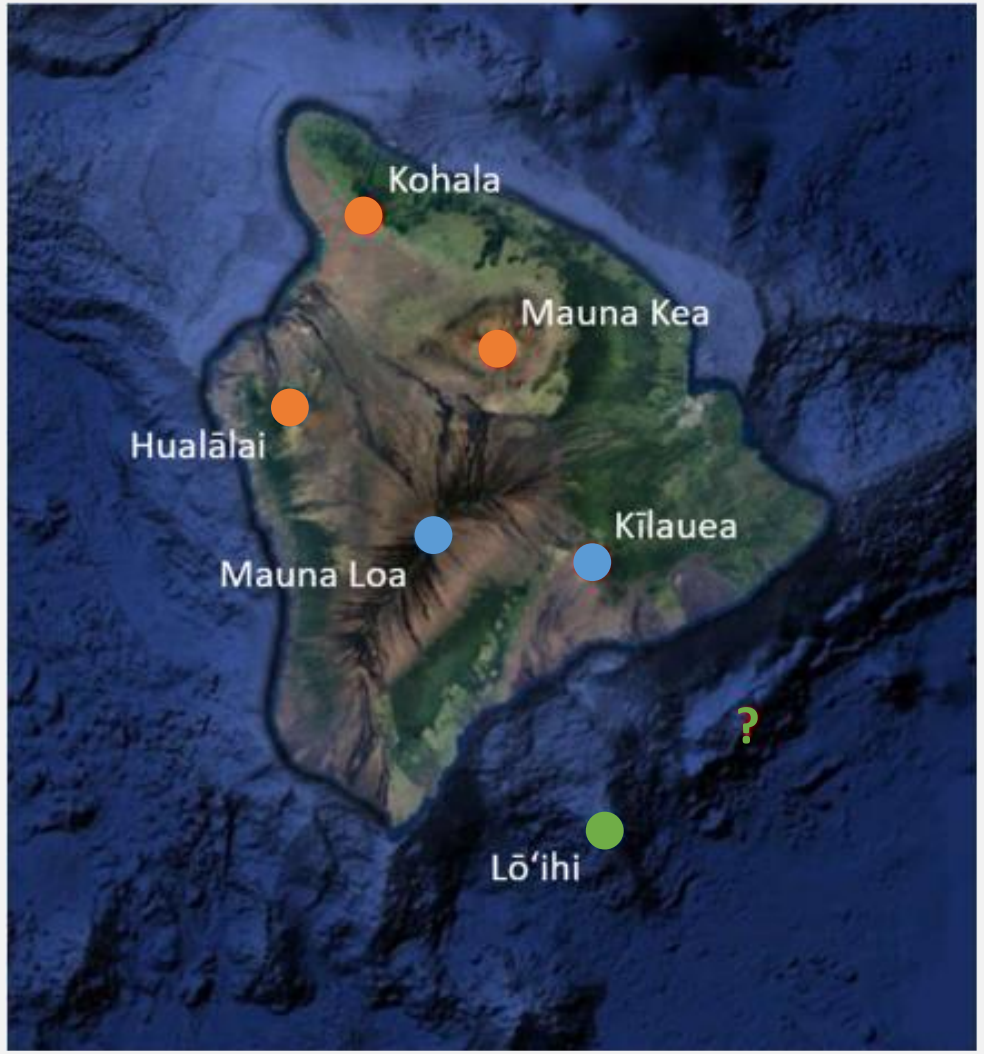


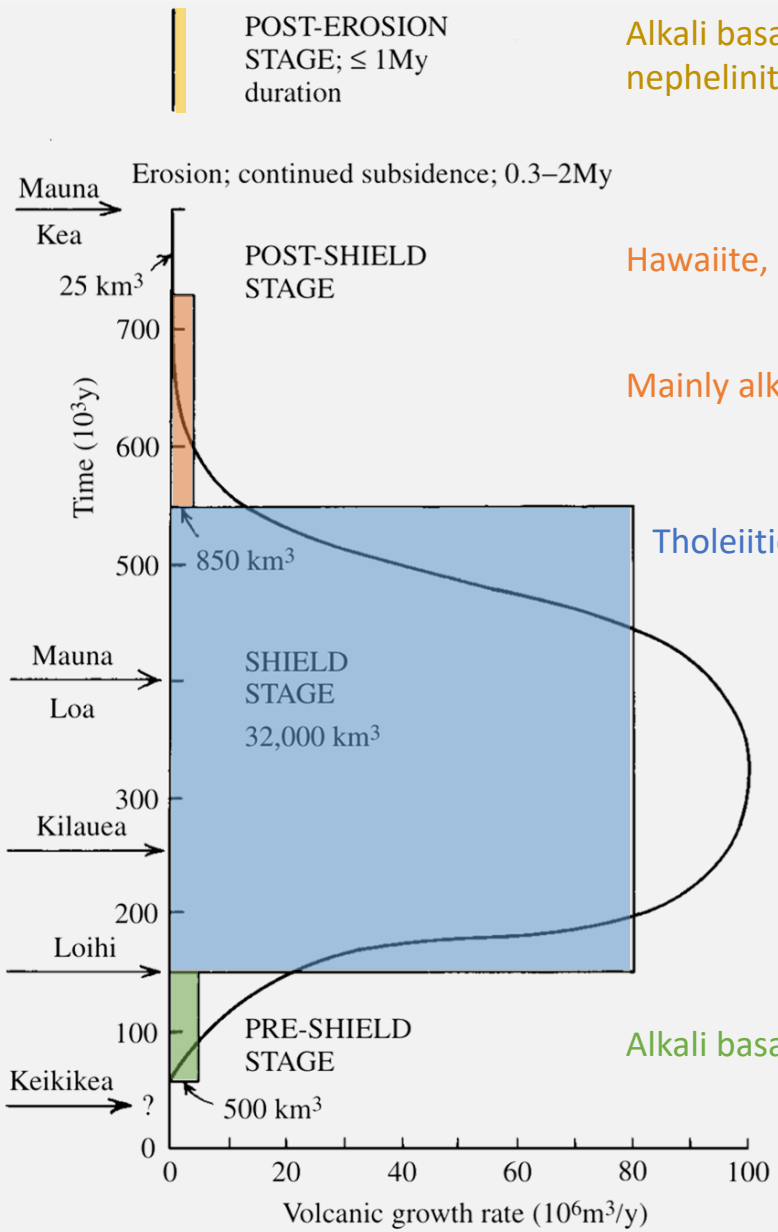


There are two near-parallel chains of age-progressive volcanism on the Hawaiian islands. The magmas of the Loa and Kea trends have different isotopic compositions, and this is linked to the isotopic composition of the mantle peridotite being melted.

margaret.hartley@manchester.ac.uk

Eruptive lifecycle of volcanoes on Hawaii



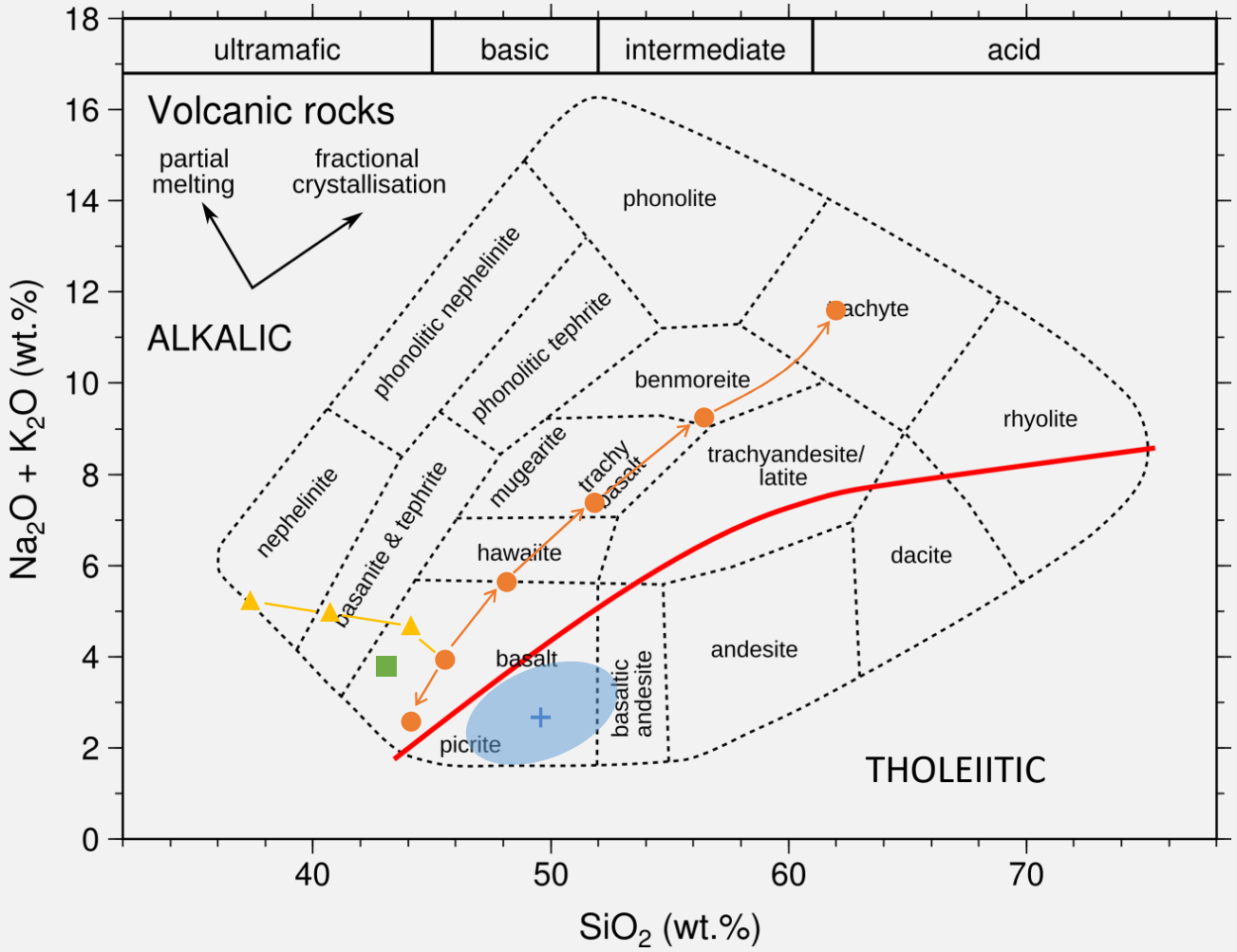


Magma volume and composition both change systematically over the volcano's lifetime

Best (2006) Fig. 13.11

margaret.hartley@manchester.ac.uk

Eruptive lifecycle of volcanoes on Hawaii: magma compositions



Post-erosional stage: Strongly silica-undersaturated alkali magmas

Post-shield stage: Alkali basalts to trachytes

Shield stage: Tholeiitic basalts

Pre-shield stage: Alkali basalts

Magma compositions span the alkali-tholeiite divide. The basalts show evidence for early olivine fractionation followed by crystallisation of cpx + plag.

Post-erosional magmas are the very last gasps of volcanism and have unusual silica-undersaturated compositions.

margaret.hartley@manchester.ac.uk

Intraplate volcanism

Oceanic intraplate volcanism: Ocean islands. Case study: Hawaii

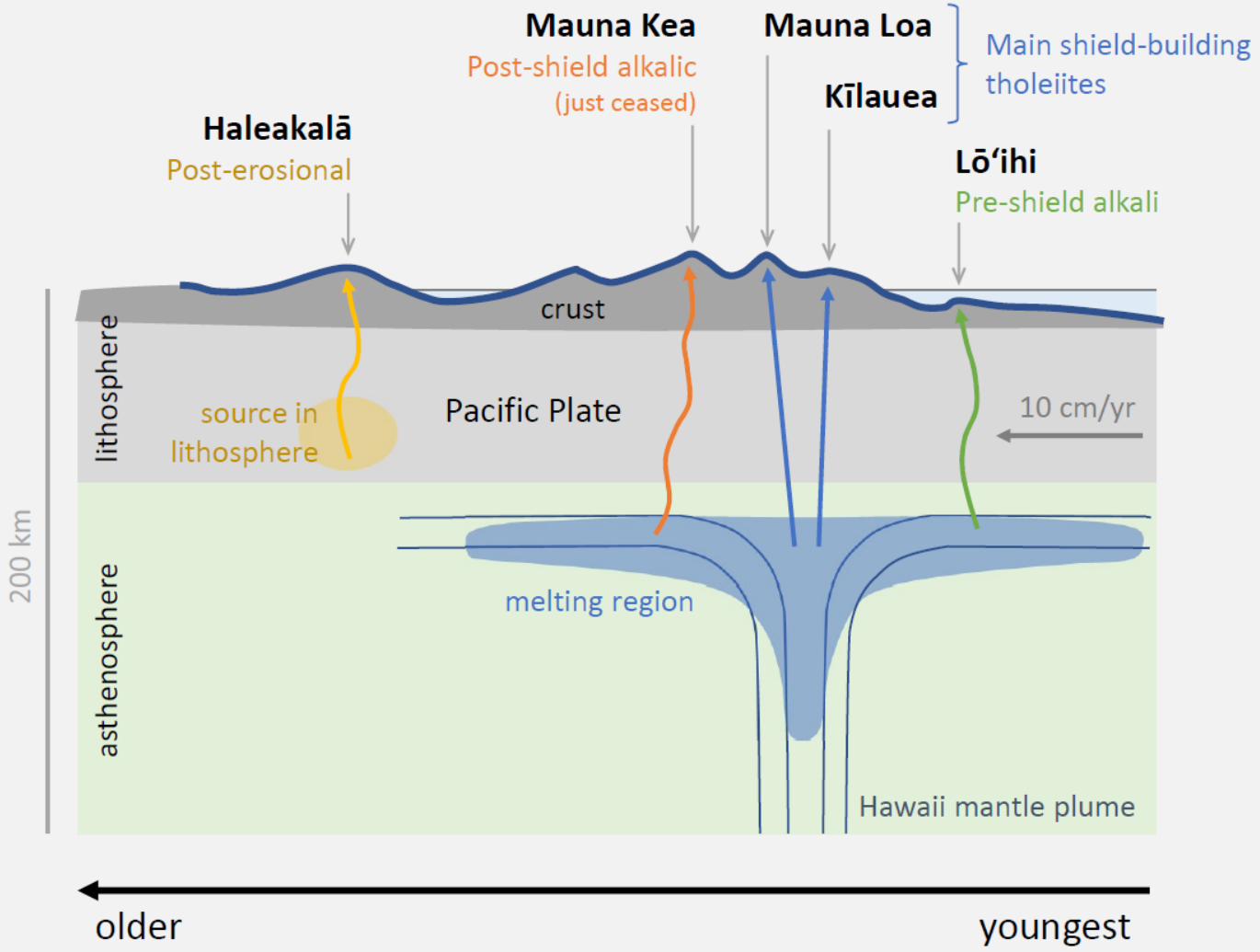
2.1 Lifecycle of Hawaiian volcanoes

2.2 Melt production over the Hawaii mantle plume



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Age progression and lifecycle of Hawaiian volcanoes is due to plate motion over the plume

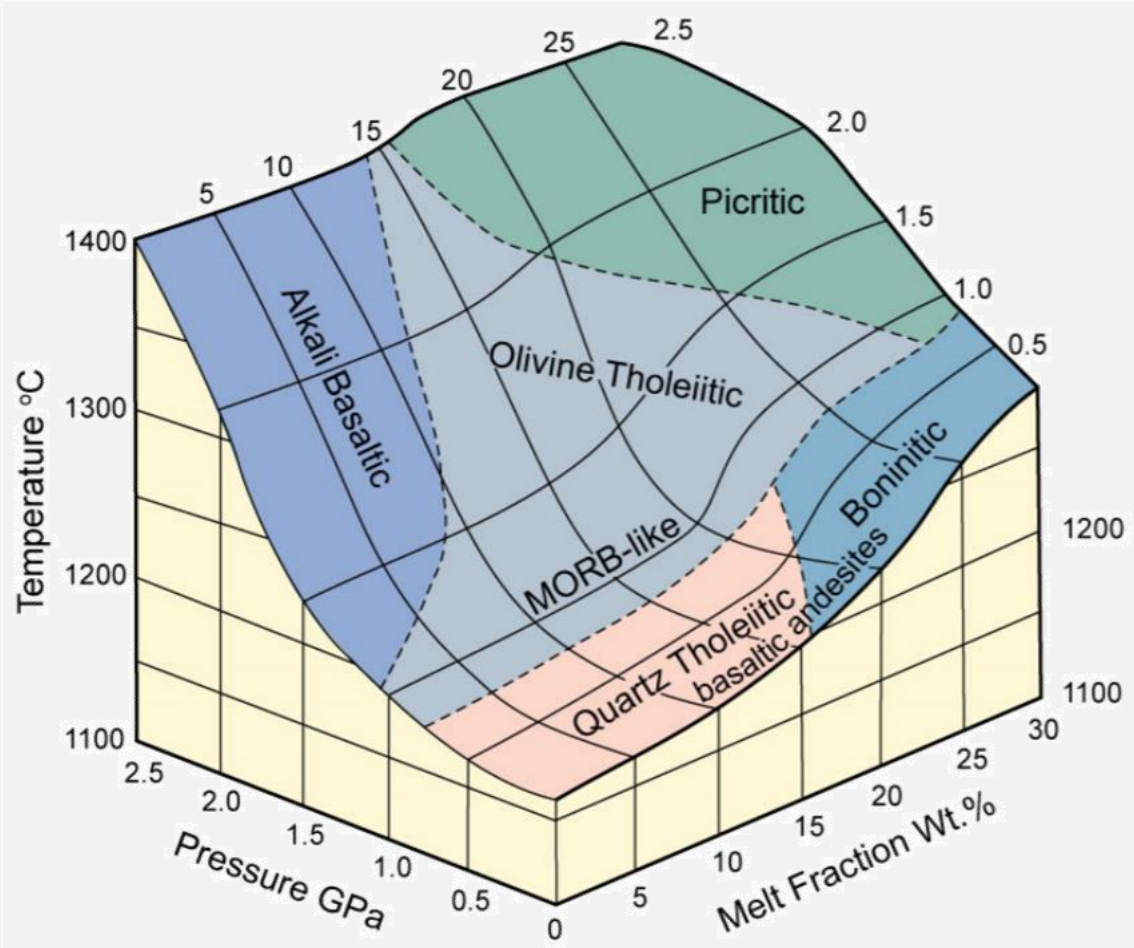


The substantial compositional differences between magmas erupted from volcanoes of differing age leads to the idea that there is a distinct life-cycle to Hawaiian volcanism.

The life-cycle of each volcano tracks the motion of the lithosphere over the plume stem.

margaret.hartley@manchester.ac.uk

What controls the primary melt composition being produced?



Primary melt composition (X) varies as a function of temperature (T), pressure (P) and the extent of partial melting (F)

Directly above the plume stem: Tholeiitic basalts

- Highest mantle temperature
- Higher degree of partial melting, F
- Melting occurs over a wide interval of pressures
- High volume of primary melt produced

At the edges of the plume head: Alkali basalts

- Lower mantle temperature
- Lower degree of partial melting, F
- Melting is confined to relatively higher pressures
- Small volume of primary melt produced

margaret.hartley@manchester.ac.uk

Summary: Time-progressive variation in melt production for a Hawaiian volcano

Pacific plate moves over the Hawaiian plume.

1. The first melts are produced when the edge of the plume starts to impinge on the lithosphere.
    - Mantle temperature begins to increase
    - Low degrees of partial melting, at high pressure, produces alkali basalts.

2. Plate motion brings the volcano over the plume centre. Mantle temperature is at its highest.
    - Higher degrees of partial melting, over a range of pressures, produces tholeiitic basalts

3. Plate motion moves the volcano over the plume tail.
    - Mantle temperature decreases
    - Low degrees of partial melting, at higher pressures, produces alkali basalts.
    - Some small-fraction melts do not erupt, but instead solidify as intrusions in the lithosphere

4. Plate motion moves the volcano away from the hotspot.
    - Return to ambient mantle temperature. Mantle melting stops.

5. Mass removal by erosion of the volcanic edifice. Intrusions that were previously frozen in the lithosphere may be re-melted by adiabatic decompression melting.
    - Very low degrees of melting produce strongly silica-undersaturated alkaline magmas with unusual compositions

margaret.hartley@manchester.ac.uk

Intraplate volcanism

You should now be able to:

- Describe the lifecycle of a Hawaiian volcano in terms of magma volume and magma composition
- Explain how time-progressive changes in volcanism relate to plate motion over the plume stem
- Qualitatively explain the factors that control the compositions of primary melts generated above different parts of a mantle plume
- Compare and contrast melt generation at mid-ocean ridges and ocean islands



margaret.hartley@manchester.ac.uk

Exam-style question

Compare and contrast the melting processes occurring at mid-ocean ridges and ocean islands. Your answer should include information on the type of mantle being melted, the conditions of melting (pressure and temperature), and the types of primary melts being produced.



margaret.hartley@manchester.ac.uk

Intraplate volcanism

3.1 A review of trace elements in igneous petrology

3.2 Trace element signatures of MORB and OIB

3.3 Do OIBs provide evidence for a compositionally layered mantle?



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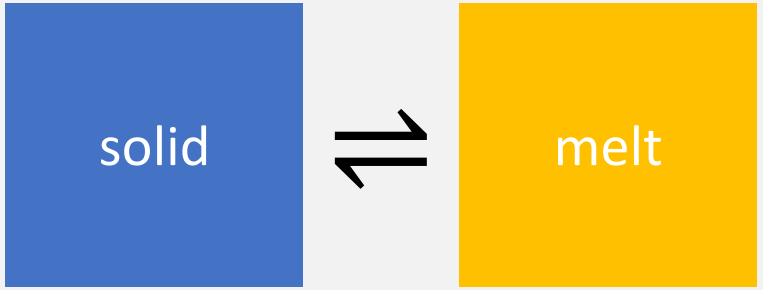
Partition coefficients

See also EART 26201 Principles of Geochemistry, Lecture 1

Trace elements have very low concentrations in magmas, measured in **ppm**

The partition coefficient is the ratio of the concentration of element *i* in mineral *j* to the concentration of element *i* in the melt at equilibrium:

Chemical equilibrium



$$D_i = \frac{C_i^{solid}}{C_i^{melt}}$$

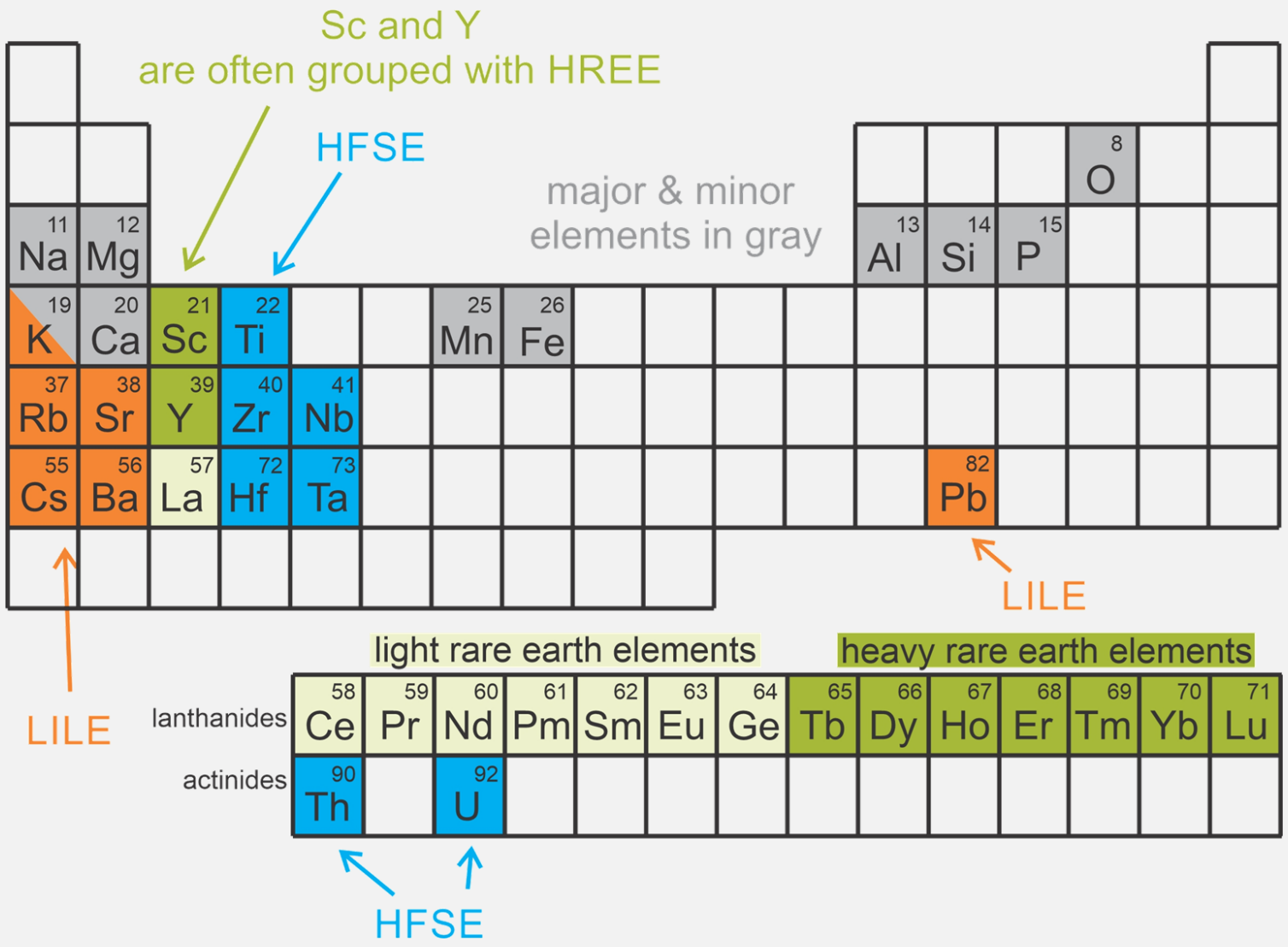
$D_i > 1$ Element i is compatible: it partitions into the solid

$D_i < 1$ Element i is incompatible: it partitions into the melt

margaret.hartley@manchester.ac.uk

Geochemist's periodic table: Types of incompatible trace element

See also EART 26201 Principles of Geochemistry, Lecture 1



Key:

- Major elements
- Light ion lithophile elements (LILE)
- High field strength elements (HFSE)
- Rare earth elements (REE)

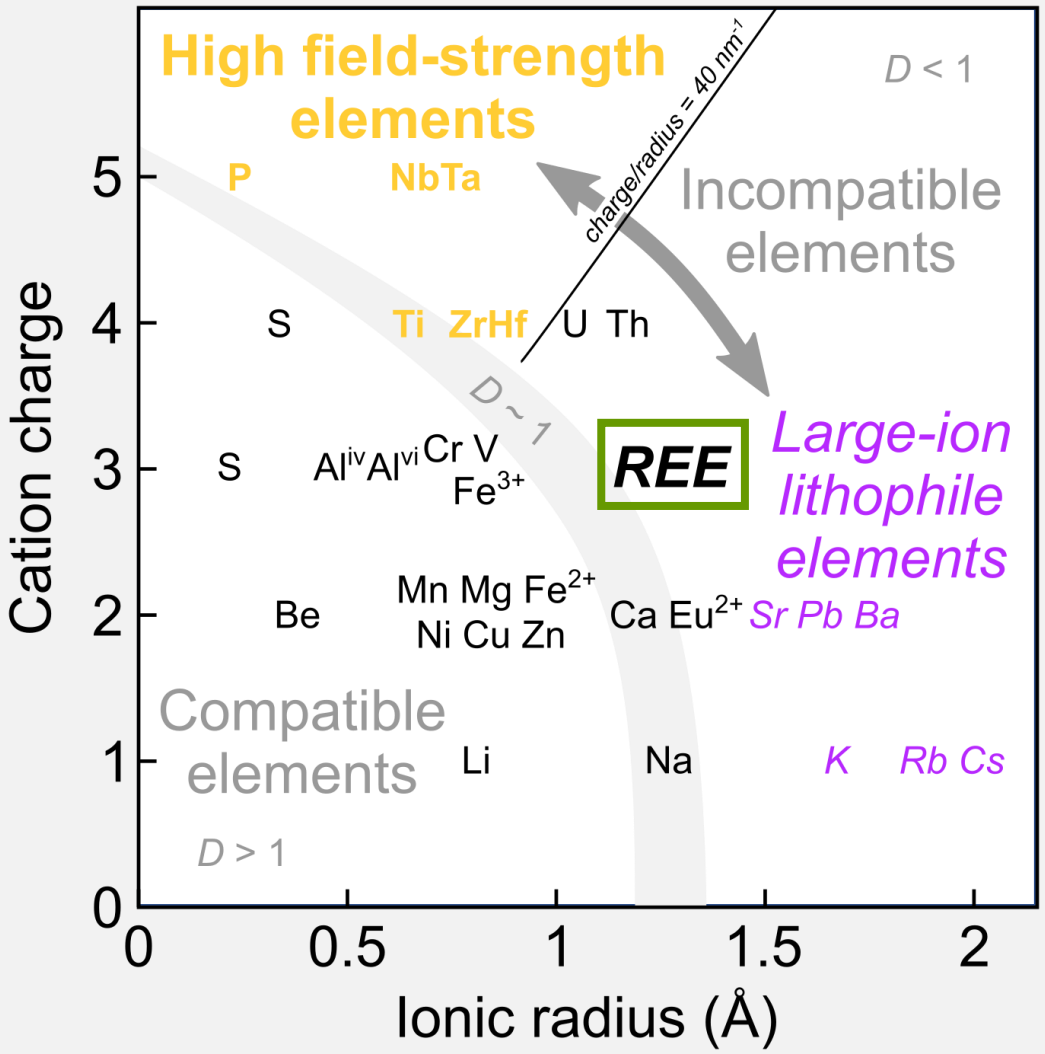
Rare Earth Elements, REE

Subdivided into light (LREE) and heavy (HREE)

margaret.hartley@manchester.ac.uk

Types of incompatible trace element

See also EART 26201 Principles of Geochemistry, Lecture 1



High field strength elements (HFSE)
Too highly charged to fit into crystal lattices
Largely immobile in aqueous fluids

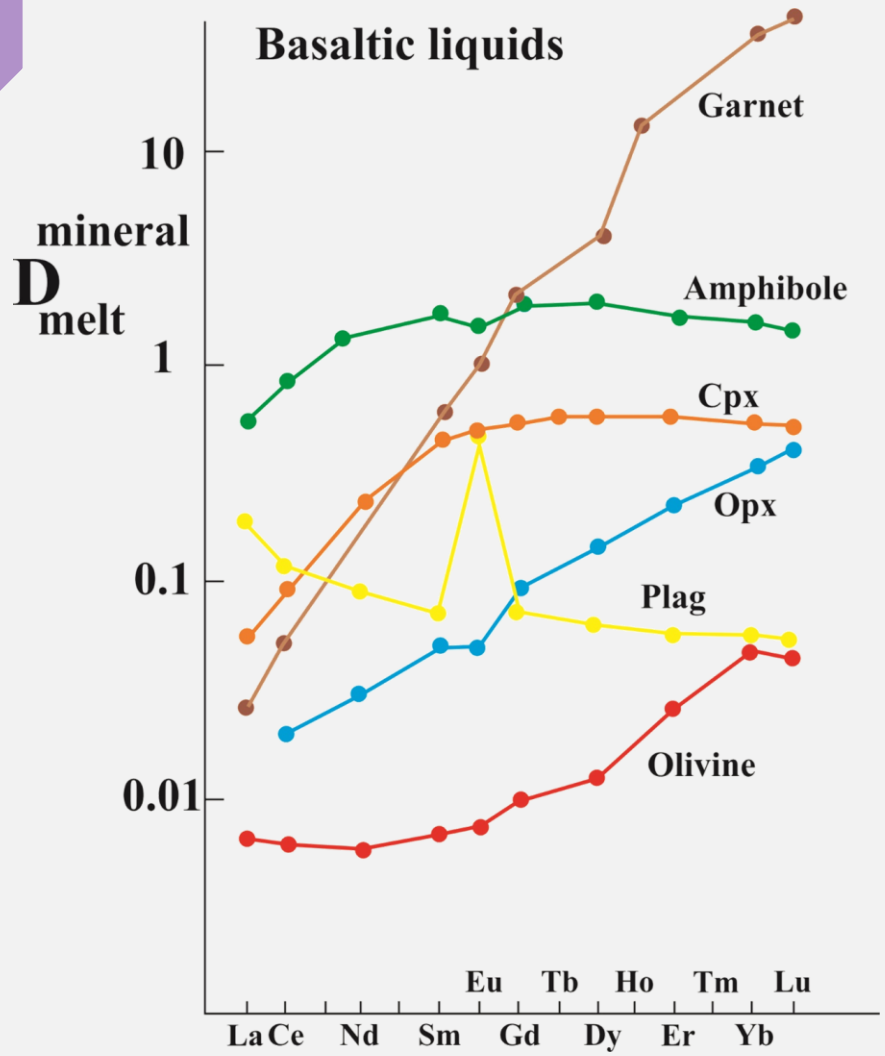
Large ion lithophile elements (LILE)
Too large to fit into crystal lattices
Highly mobile in aqueous fluids

Rare earth elements (REE)
Show related behaviour during key geological processes like melting and crystallisation

We will revisit LILE and HFSE in Week 8

margaret.hartley@manchester.ac.uk

REE partition coefficients in minerals



The rare earth elements (REEs) exhibit very coherent partitioning behaviour in silicate minerals. This makes them very powerful geochemical tracers.

LREEs are incompatible in most silicate minerals.

HREEs are highly compatible in **garnet**.

- Therefore, partial melts of garnet peridotite mantle are depleted in HREE.

Eu^{2+} is more compatible in plagioclase than any other REE.

- If plagioclase has crystallized from a melt, then the residual liquid will have a negative Eu anomaly.

margaret.hartley@manchester.ac.uk

Intraplate volcanism

3.1 A review of trace elements in igneous petrology

3.2 Trace element signatures of MORB and OIB

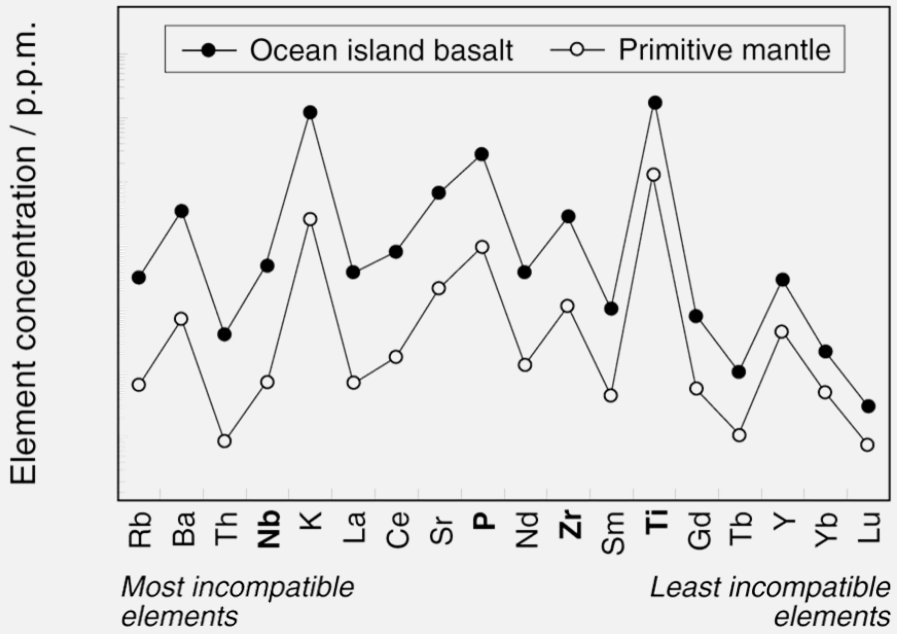
3.3 Do OIBs provide evidence for a compositionally layered mantle?



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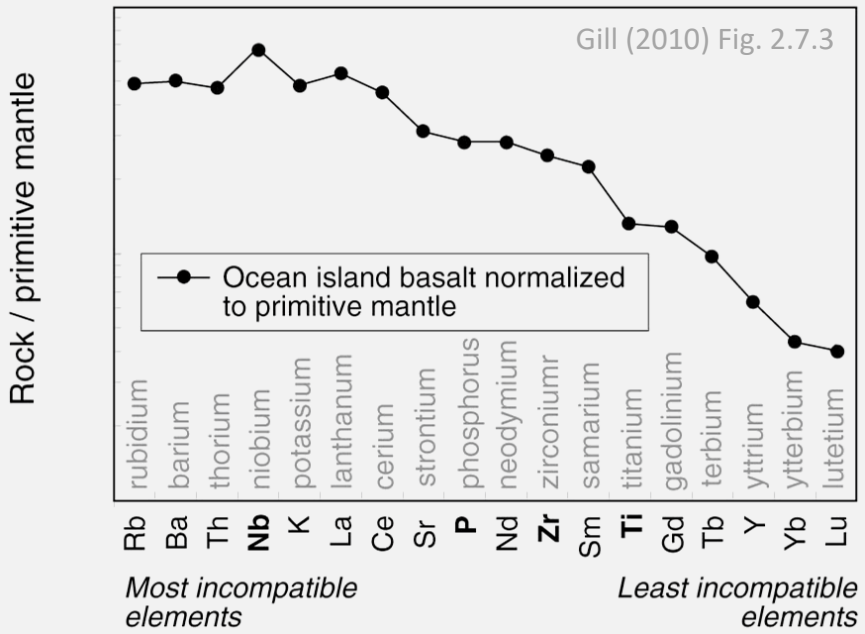
Trace elements: Geochemical fingerprints of MORB and OIB

See also EART 26201 Principles of Geochemistry, Lecture 1



y axis: Absolute elemental concentrations.

Oddo-Harkins rule: elements with even numbers are more abundant



y axis: Element concentrations normalized to an appropriate geochemical reservoir.

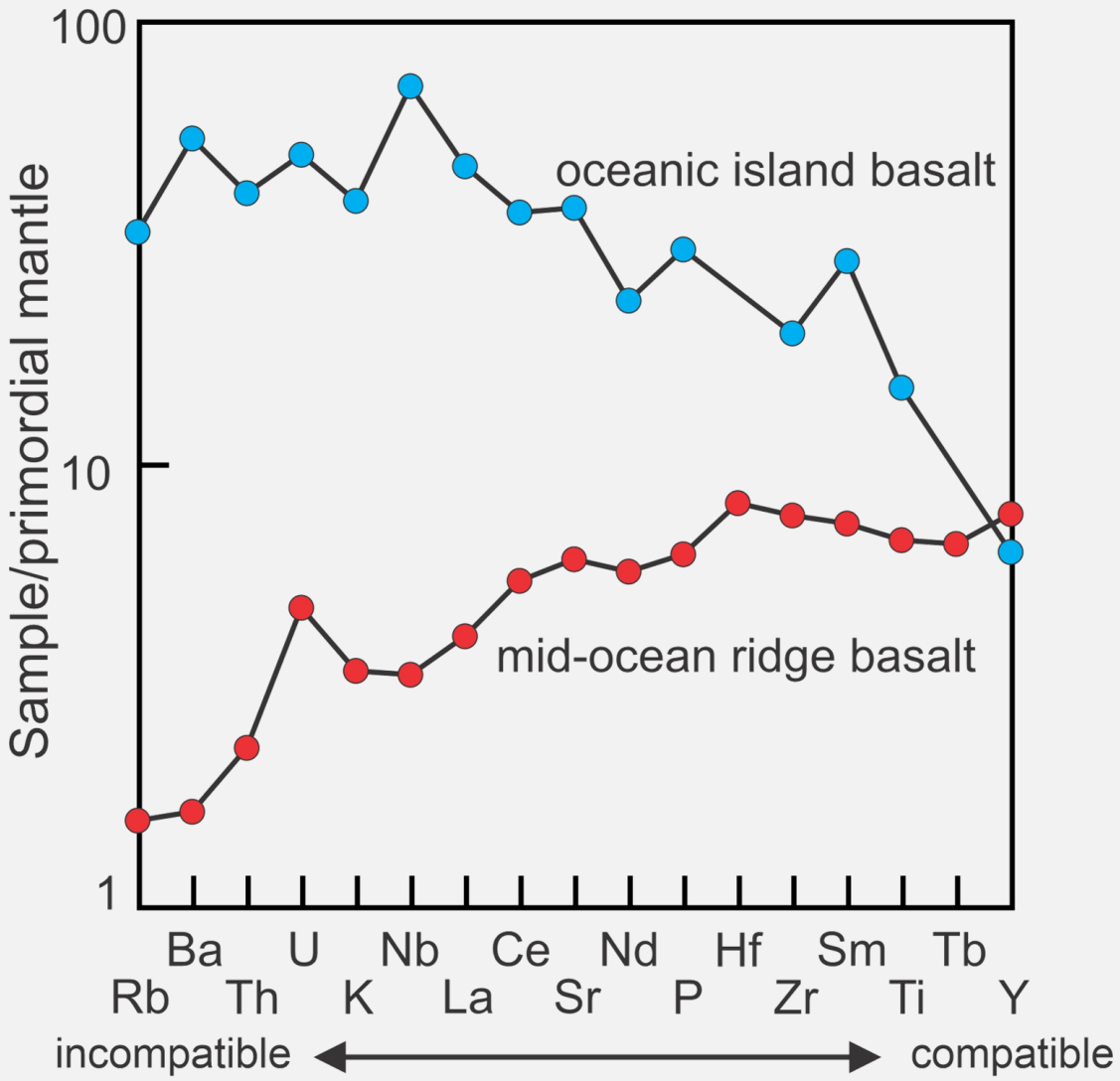
Oddo-Harkins effect is removed; easier to compare different rock or magma types

Common geochemical reservoirs for normalization:

- Primitive mantle (PM)
- Depleted MORB mantle (DMM)
- Chondritic uniform reservoir (CHUR)

margaret.hartley@manchester.ac.uk

Trace elements: Geochemical fingerprints of MORB and OIB



OIB:

- Steep negative gradient
- Enriched in incompatible trace elements
- Depleted in compatible trace elements, especially HREE

MORB:

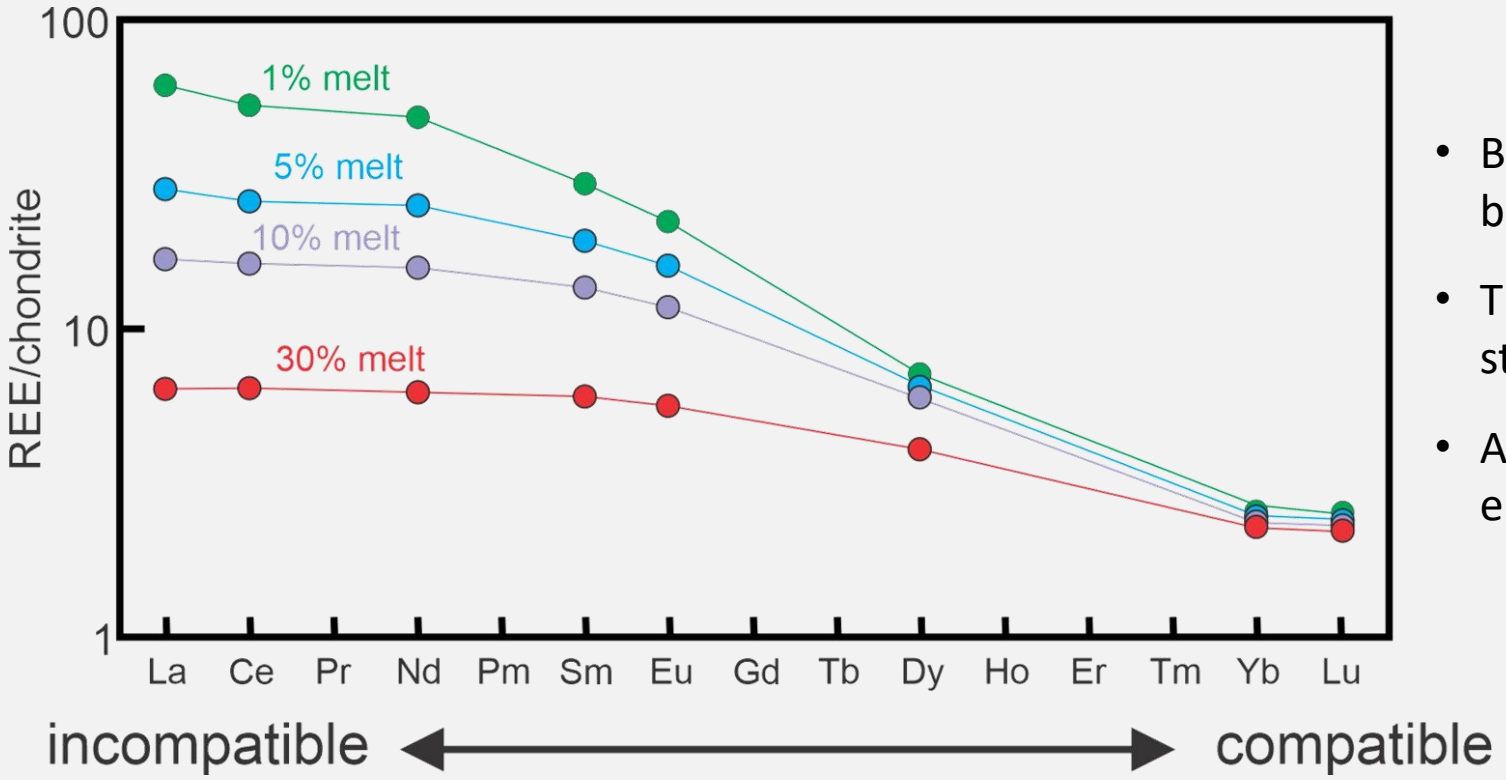
- Positive gradient
- Strongly depleted in incompatible trace elements
- Depleted in all trace elements compared to OIB

How could these differences be generated?

1. Different degrees of partial melting of the mantle
2. Different mantle source compositions

margaret.hartley@manchester.ac.uk

Different degrees of partial melting can't explain the differences between MORB and OIB

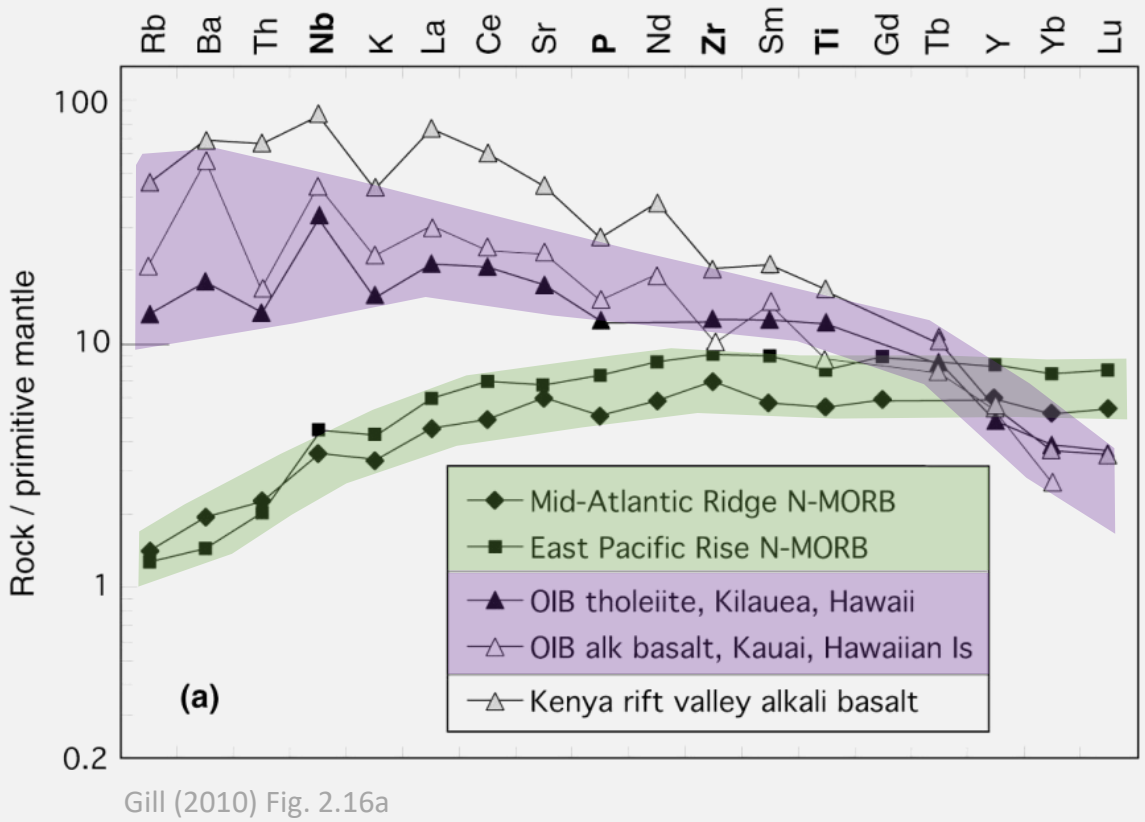


- By definition, incompatible elements will always be enriched in the melt compared to the solid.
- The most incompatible elements are the most strongly enriched in the melt.
- As partial melt fraction increases, the degree of enrichment decreases.

- All these plots have a similar shape because are all derived by melting the same mantle type (garnet lherzolite)
- The differences in gradient are created by different degrees of melting

margaret.hartley@manchester.ac.uk

Different mantle sources can explain the compositional differences between MORB and OIB



Gill (2010) Fig. 2.16a

MORB tholeiites:

- Positive gradient
- Depleted in incompatible trace elements

Hawaii OIB (tholeiites and alkali basalts):

- Steep negative gradient
- Enriched in incompatible trace elements
- Depleted in HREE: indicates melting of garnet-bearing mantle

The differences between MORB and OIB cannot be created by partial melting of the same type of mantle.

MORB and OIB must be partial melts of different kinds of mantle: different composition and/or different mineralogy

margaret.hartley@manchester.ac.uk

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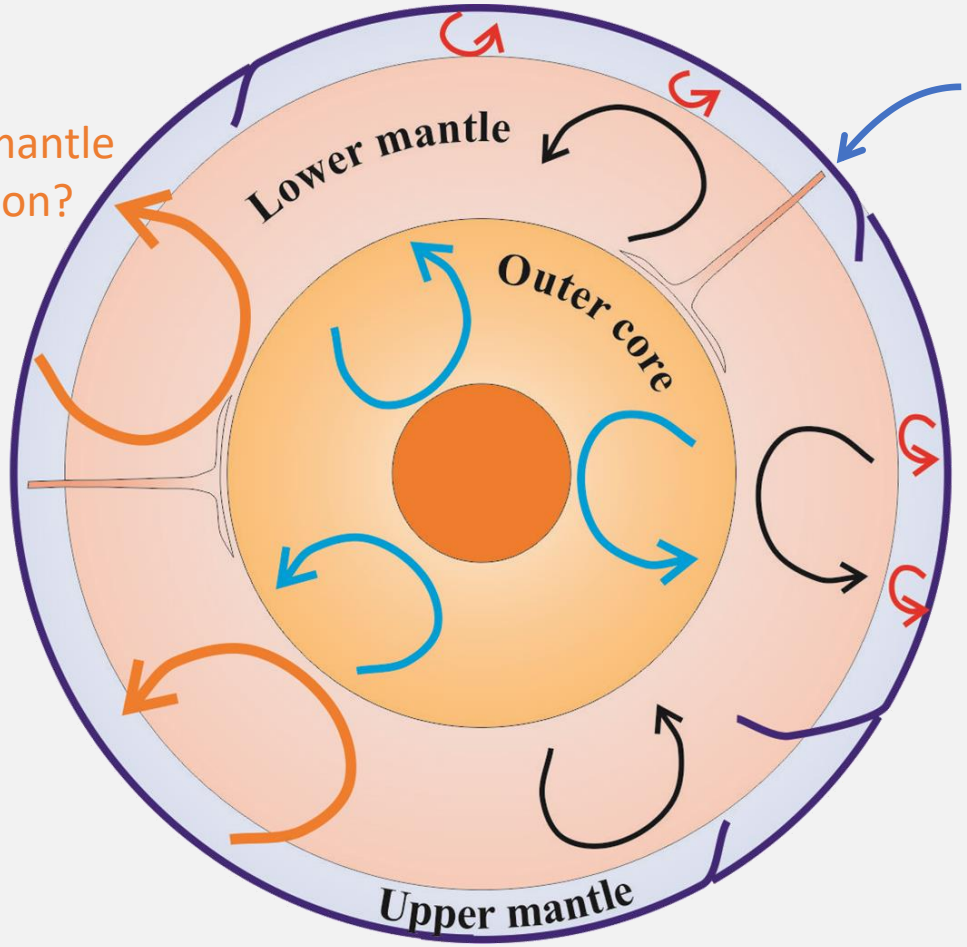
3.3 Do OIBs provide evidence for a compositionally layered mantle?



margaret.hartley@manchester.ac.uk

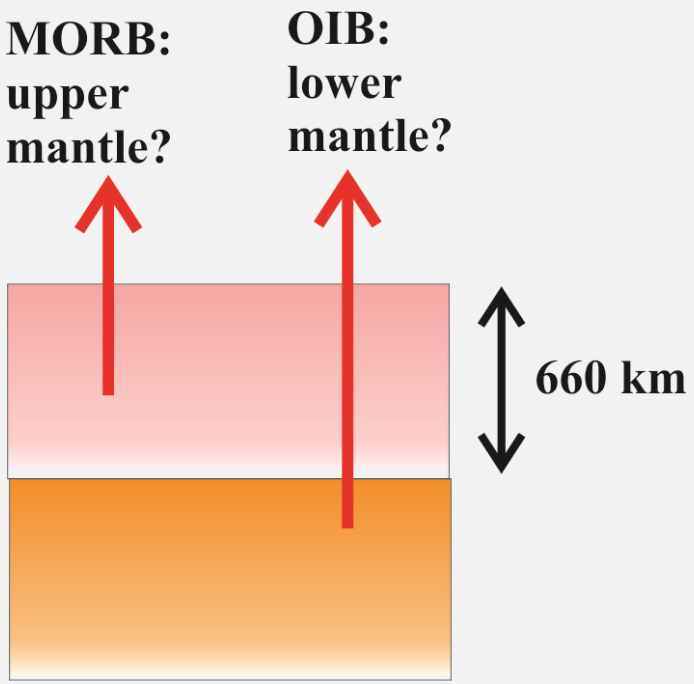
Mantle convection: whole mantle or layered mantle?

Whole mantle convection?



Ocean islands above plumes

Layered mantle convection?

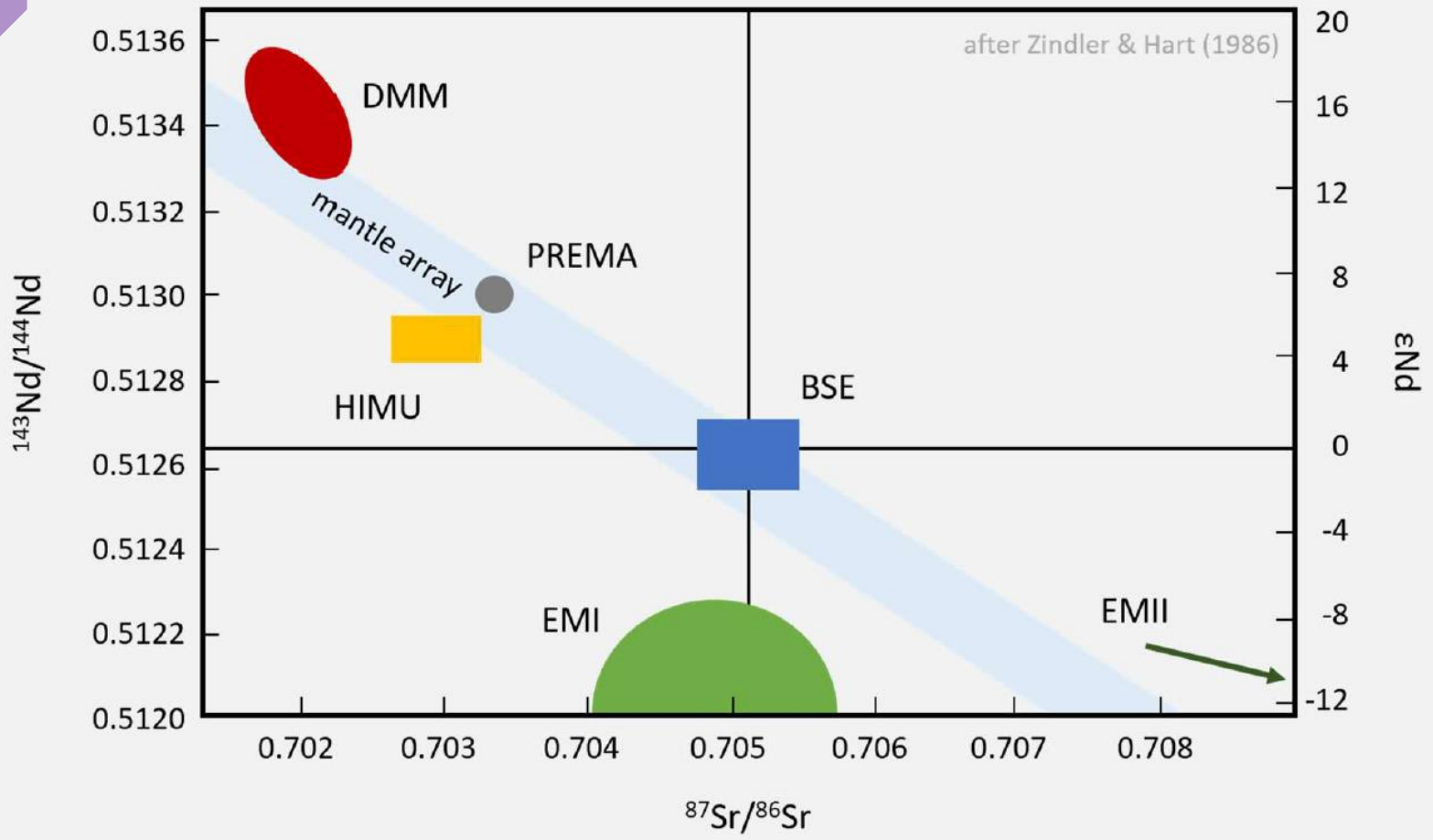


Compositional differences between MORB and OIB suggest melting of two different mantle reservoirs

margaret.hartley@manchester.ac.uk

Different mantle reservoirs have distinctive isotopic signatures

See also EART 26201 Principles of Geochemistry, Lecture 2



DMM = depleted MORB mantle

BSE = Bulk Silicate Earth. Also referred to as **PM**, Primitive Mantle

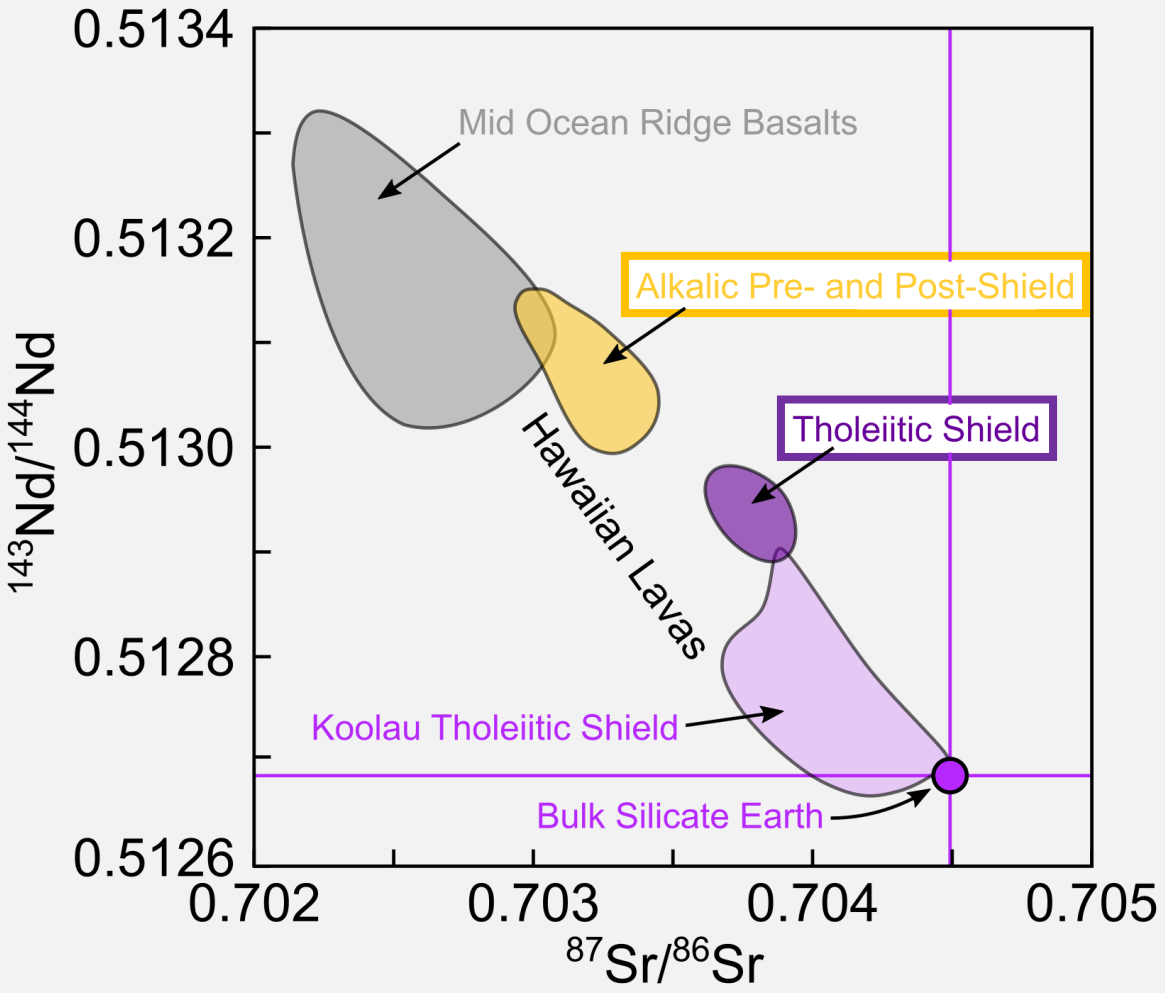
EM = Enriched Mantle. Two distinct types: EM1 and EM2

PREMA = PREvalent MAntle

HIMU = HIgh MU (μ), where μ is the $^{238}U/^{204}Pb$ ratio

margaret.hartley@manchester.ac.uk

Isotopic signatures of Hawaiian basalts consistent with melting in the lower mantle



margaret.hartley@manchester.ac.uk

Intraplate volcanism

You should now be able to:

- Define compatible and incompatible trace elements
- Identify the trace element signatures of typical MORB and OIB on a normalized trace element plot
- Describe the compositional differences between MORB and OIB, and relate these to the composition and structure of the Earth's mantle



margaret.hartley@manchester.ac.uk

Intraplate volcanism

Continental intraplate volcanism

4.1 Large igneous provinces



margaret.hartley@manchester.ac.uk

Large igneous provinces (LIPs)

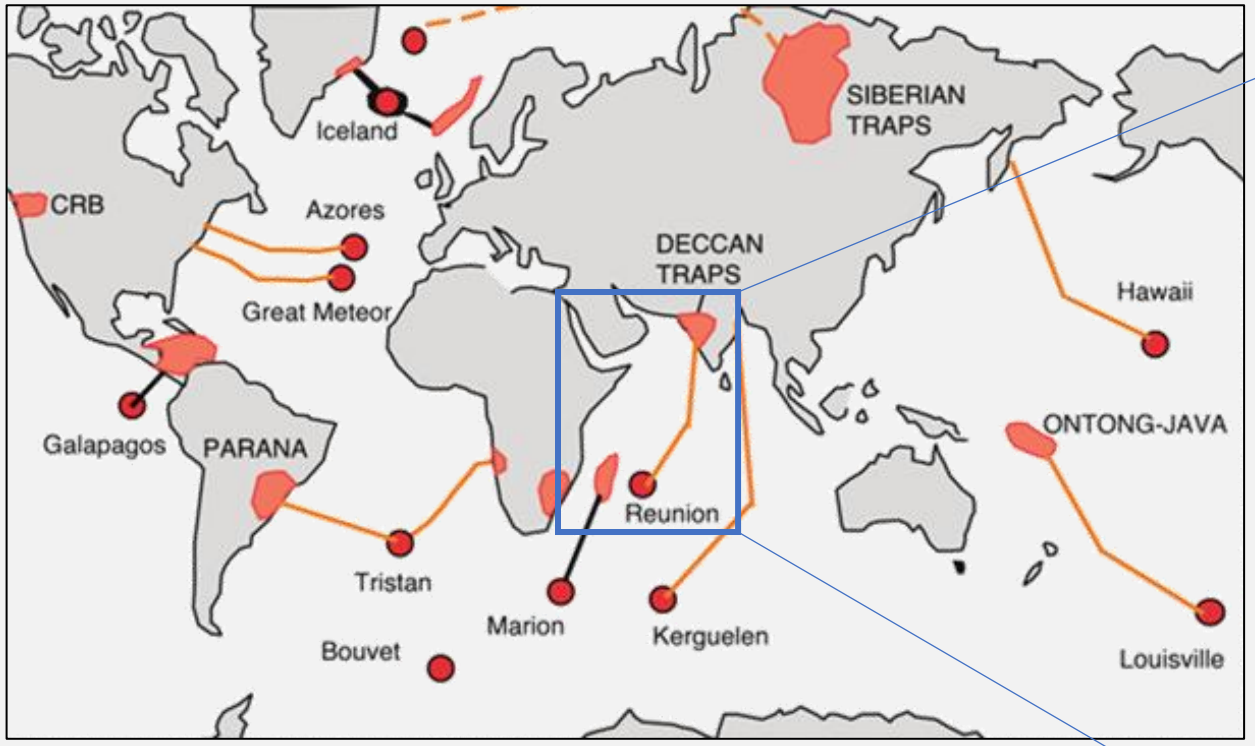


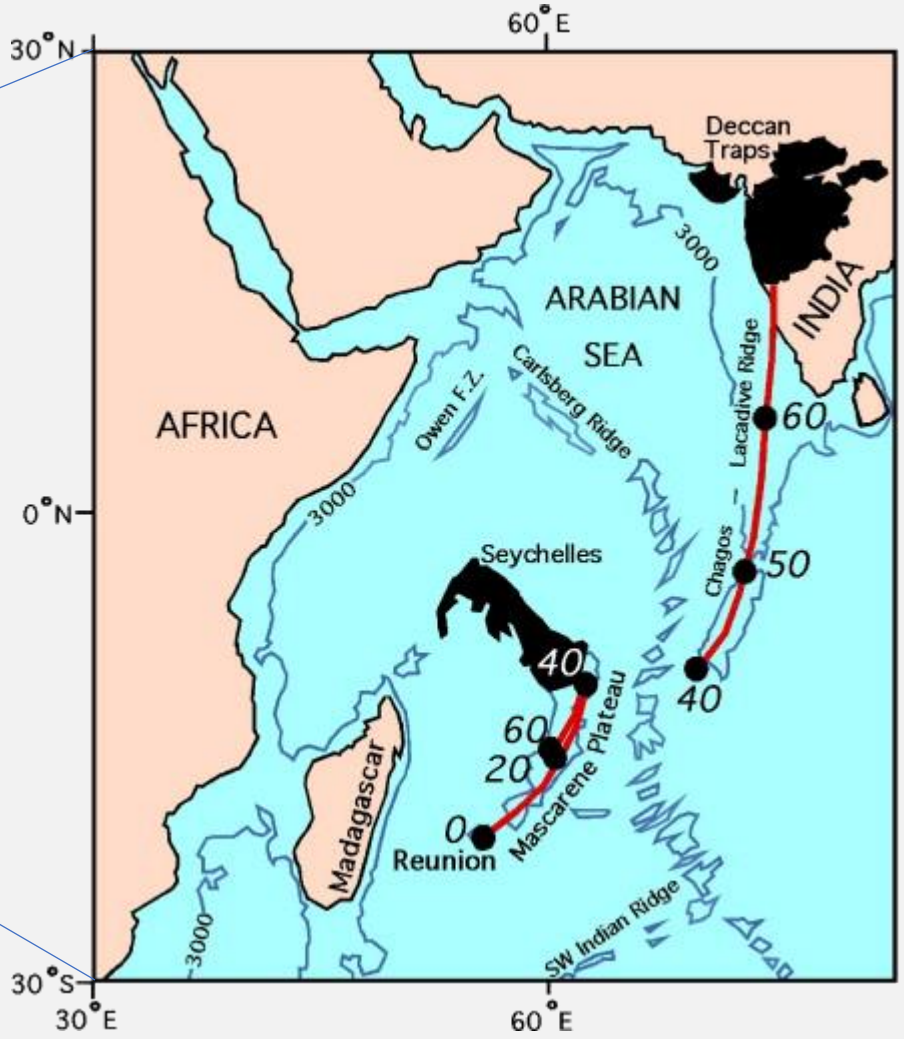
- LIPs are voluminous outpourings of magma erupted over a relatively short period of geological time (~1 Ma or less)
- The formation of many LIPs is attributed to mantle plumes
- Some LIPs are associated with continental break-up (e.g. CAMP, Parana-Etendeka)
- LIPs erupted through continental lithosphere are often referred to as continental flood basalts (CFB)

PDD = Pripyat-Dneipr-Donets
CAMP = Central Atlantic Magmatic Province

margaret.hartley@manchester.ac.uk

Many LIPs can be linked to known hotspots

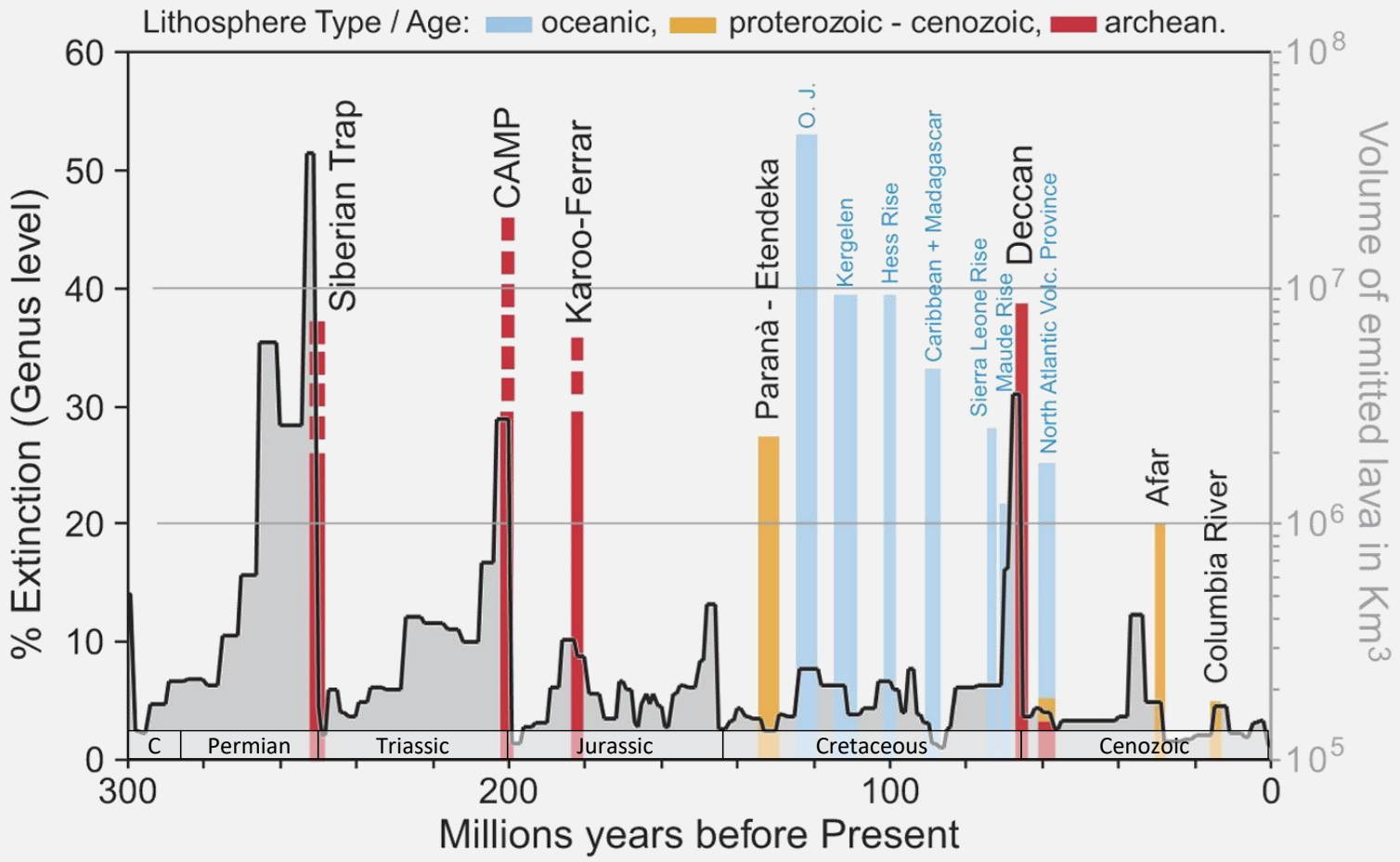




Some LIPs have hotspot trails linking them with modern-day ocean islands. This is consistent with the hypothesis that LIP magmatism is associated with mantle plumes

margaret.hartley@manchester.ac.uk

Some LIPs are associated with mass extinctions

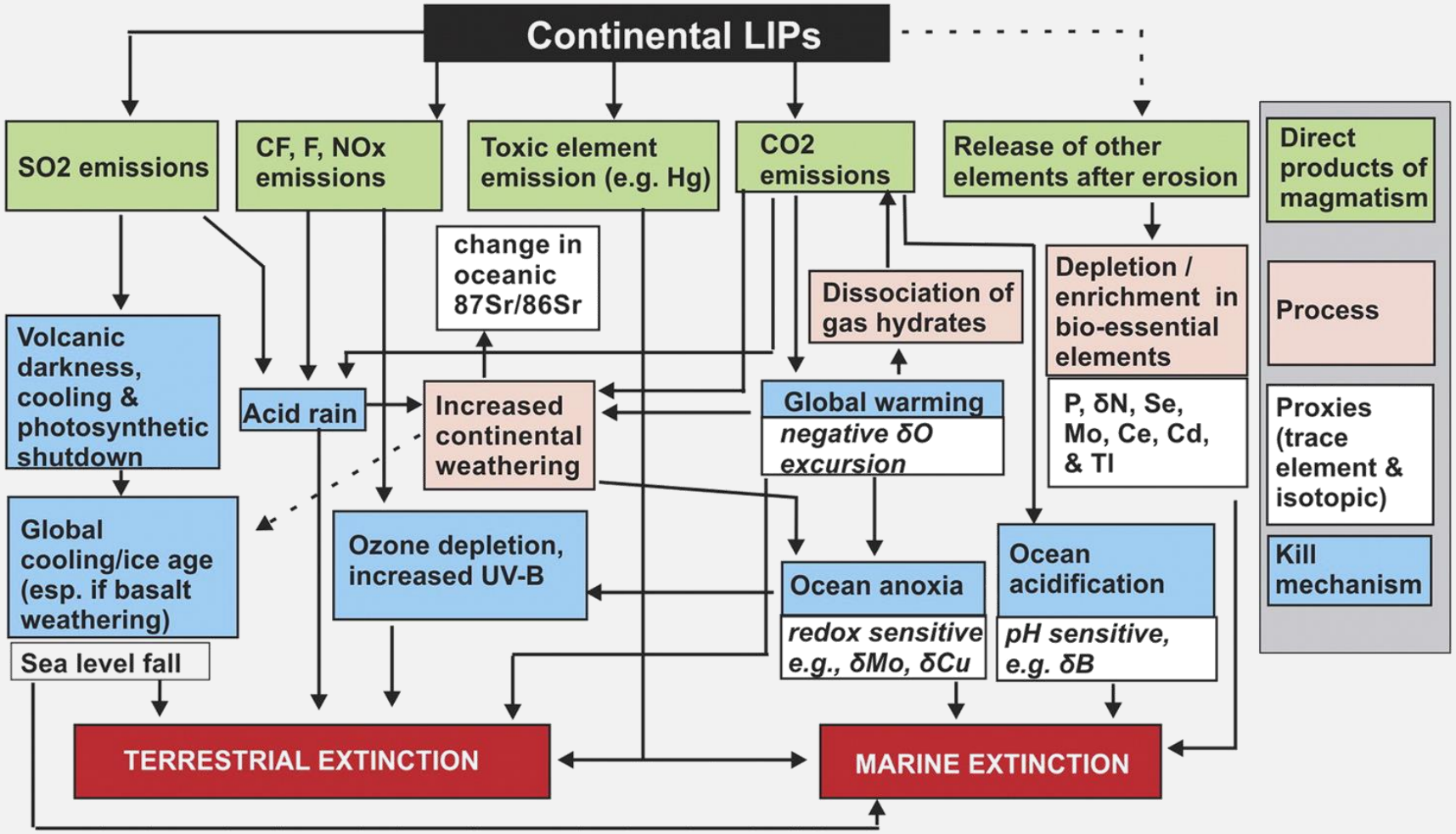


Possible drivers of environmental stress:

- CO_2 emissions: greenhouse gas
- SO_2 emissions: greenhouse gas; acidification of oceans, rain and soils
- Sulfate aerosols: ozone destruction leading to intense UV radiation
- Ash emissions: block incoming solar radiation
- Emissions of toxic metals

margaret.hartley@manchester.ac.uk

Environmental effects of continental LIPs

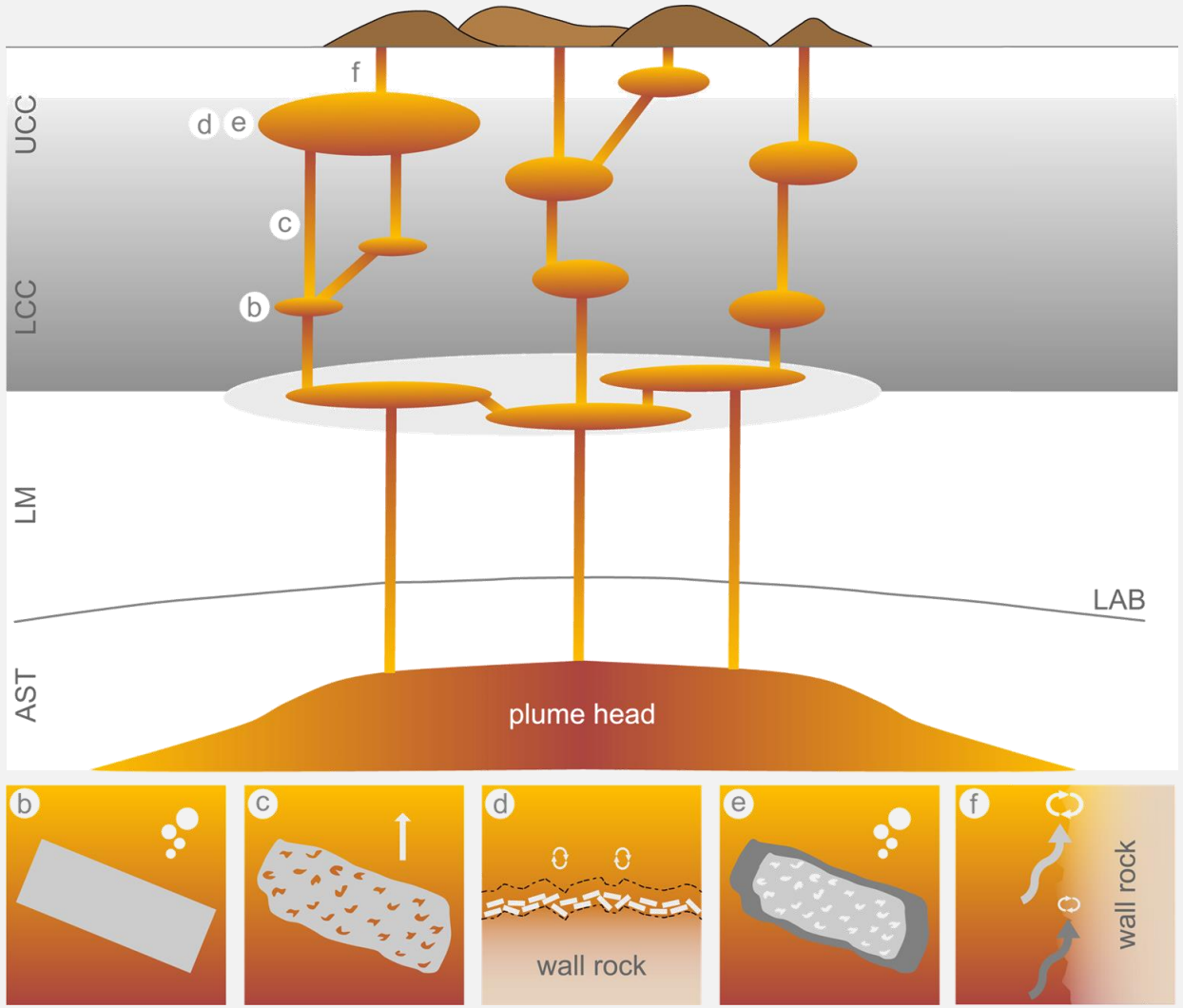


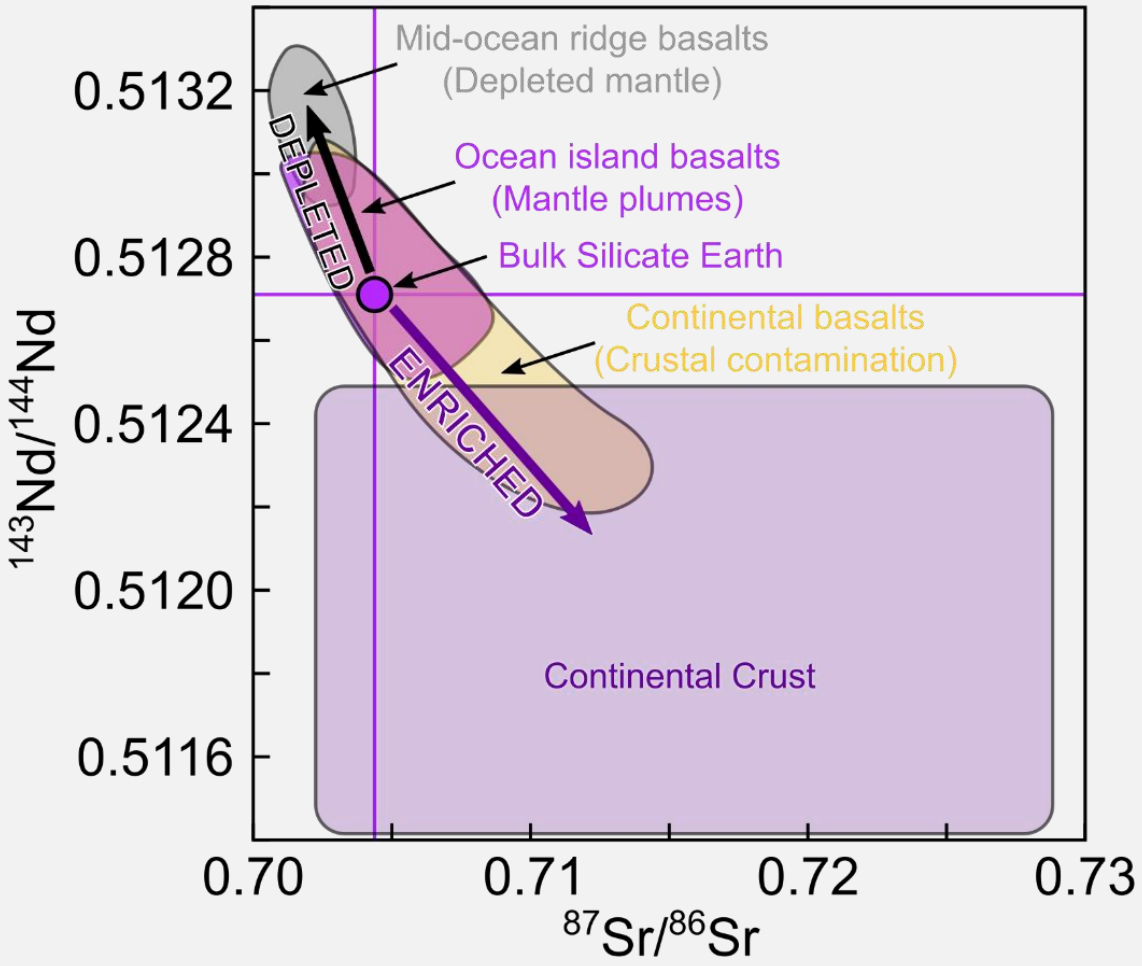
Flow diagram for oceanic LIPs: Ernst and Youbi (2017) Fig. 5

margaret.hartley@manchester.ac.uk

Continental LIPs erupt through thick lithosphere and assimilate continental crust

We will revisit crustal melting in weeks 8 and 9





margaret.hartley@manchester.ac.uk

Intraplate volcanism

You should now be able to:

- Describe the geological correlation between large igneous provinces and mass extinctions
- Compare and contrast oceanic and continental intraplate volcanism



margaret.hartley@manchester.ac.uk